GEORABIES: MOBILE ANDROID-BASED ANIMAL BITE REPORTING AND VACCINATION LOCATOR WITH INTEGRATION OF GIS IN LOS BAÑOS

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In Partial Fulfillment of the Requirements for the Degree of
Bachelor of Science in Information Technology
Specialized in Web and Mobile Application Development

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Mr. & Mrs. Abas

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ABSTRACT

Rabies cases are still one of the public health concerns because of the deadly virus that can infect the human brain through bites from a rabid animal, and the lack of accessible reporting tools that causes underreporting of animal bite cases. It is important to make a decision and take action if a system is in place to monitor animal cases for the community's safety and to raise awareness. The GeoRabies application aims to assist users in reporting animal bite incidents by providing basic information about the incident, such as name, age, date of the incident, type of animal (dog, cat, other), animal category (stray, wild, pet), and locating nearby clinics that are available in the community. It integrates a Geographic Information System (GIS) for mapping and clinic visualization, and enables users to identify available and nearest clinics for rabies vaccination. The researchers used the ISO 25010 software quality model for system evaluation. The app includes animal bite reporting, a mapping module, a clinic locator, an animal bite report generator, and an admin dashboard. Findings revealed that the system is perceived to meet the essential software quality characteristics de ned under the ISO 25010 framework to a great extent. High mean scores across most quality attributes indicate that the system provides effective, reliable, and supported rabies prevention efforts.

Keywords: Geographic Information System (*GIS), ISO 25010, Mobile-Based Android Application, Animal Bite Reporting, Vaccination Locator, Incident management*

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DEFINITION OF TERMS

Expo Go	A mobile app that lets you run and test React Native applications on your phone without installing them separately. Used to preview and debug the georabies application projects during development quickly.
Firebase Database	A cloud-based backend platform that offers mobile applications scalable and real time database services. in contrast to local databases, makes it possible for data to be safely saved, synchronized, and accessed over the internet, facilitating easy data exchange between users and devices.
Figma	An online tool used for designing and creating user interfaces, websites, and app layouts. It is used to make designs, prototypes, and visualize the Georabies application.
Geographic Information System (GIS)	A computer based system that captures, stores, analyzes, and displays data linked to specific locations on a map. Used to visualize geographic information, identify patterns.
Geographic Tagging	The process of adding location details like coordinates, addresses, or place names to data. Used to show in the map where something is located like animal bite clinics
Hardware	This refers to the physical parts of a computer, such as the processor, memory, storage, and devices you use to input or view information.
Open Street Map	An collaborative, open-source mapping project that provides freely accessible geographic data used to create interactive maps on websites and apps. It is used to display maps, add markers, and see the location of animal bite clinics.
Post-Exposure Prophylaxis (PEP)	A medical treatment given after being exposed to a disease, such as rabies, to prevent infection.
Pre-exposure prophylaxis (PrEP)	A preventive treatment given before exposure to a disease, such as rabies, to reduce the risk of infection.
Visual Studio Code	A free code editor used for writing and editing programs and scripts. Which is also used by the researchers to create, test, and debug software efficiently.

CHAPTER 1

INTRODUCTION

Project Context

Mobile applications are increasingly used in the current digital era to address problems in public safety, healthcare, and education. Many disadvantaged people's inability to access high-quality information limits their opportunities. The lack of easily accessible vaccination programs in rural regions and the delayed reporting of animal bite incidents are two of the main obstacles to rabies control. Many victims are not aware of the correct post-exposure procedures, which frequently leads to avoidable deaths. Global public health is still at risk from rabies, a viral zoonotic illness that causes progressive and deadly inflammation of the brain and spinal cord

An estimated 59,000 people die from rabies each year across more than 150 countries, with 95% of incidents taking place in Asia and Africa. This sure is likely a significant underestimate due to underreporting and imprecise estimates. The impoverished in rural areas bear a disproportionate amount of the disease's burden, with children under the age of 15 accounting for about half of all cases. (World Health Organization - WHO).

The poor in Asia and Africa, the two continents with the highest global rabies burden, most benefit from elimination. The well-known shortage of reliable human and animal surveillance data now hinders the effectiveness of rabies prevention and control campaigns. (Lechenne et al. 2021). Given the high number of rabies-related deaths in Asia compared to other regions, it is determined that this region is the most affected by the virus. As a result, it indicates that there are a lot of rabid animal in Asia, and it is recommended that

the WHO, governments, and other organizations carry out their duties and implement certain measures to stop this virus. (Bilal A.)

The purpose of this project is to create an Android mobile application for reporting animal bites and locating rabies vaccinations. Users can find local immunization facilities, report bite events, and access information on rabies treatment and prevention. Using mobile technology, this approach aims to reduce rabies-related deaths by ensuring prompt reporting and access to vaccinations.

The goal of this project is to help people living in communities, especially those in areas with the highest rates of rabies exposure and the fewest healthcare options. Additionally, it facilitates incident reporting and the finding of neighboring clinics for healthcare practitioners.

Project Objectives

The main objective of this research is to develop the GeoRabies mobile application, an Android-based animal bite reporting for animal bite victims and vaccination locator that helps victims navigate to the nearby clinics with the integration of GIS in Los baños. This research specifically aims to:

1. Develop a GIS-integrated mobile application which allows users to locate nearby clinics offering bite-related treatments and access direction.
2. Create a centralized reporting system that enables users to report animal bite incidents.
3. Evaluate the developed system using ISO 25010 and conduct a user acceptance evaluation based on the technology Acceptance Model (TAM)

Project Purpose

To address this issue, a systematic approach to reporting animal bites, securing vaccination resources, and informing the public about preventive measures is necessary. Rabies is a preventable but deadly disease that still poses a threat to public health in many places. The goal of this project is to provide a rabies reporting and vaccination system for Android-based mobile applications. Users can report animal bites and access information on rabies prevention and treatment. The project intends to raise community awareness and improve rabies case management by offering a simplified platform. The project's primary beneficiaries are listed below.

Individuals (persons bitten by animals). The primary beneficiaries this project are persons who are bitten by. They could use the mobile application to file a bite incident report, as well as a mapping system that helps users navigate to the clinic for immediate treatment.

Community Members. Residents in Los Baños could benefit from access to reliable information to reduce misconceptions about the disease and raise awareness of rabies prevention through informative resources

Local Government Units (LGUs). This project monitored dog bite cases, identified barangays with high and frequent animal bites, determined the type of animal, the victim's age and gender, and provided overall analytics to visualize and present data in a way that makes complex information easier to understand. These features helped close gaps in the underreporting of animal bite incidents and enabled LGUs to plan and implement programs, especially in cities with high rabies case rates.

Scope and Limitations of the Study

This project focused on developing an Android-based mobile incident reporting system for animal bites that integrates GIS, specifically in the Los Baños area. It aims to provide users who have been bitten by an animal with directions to seek immediate treatment and first aid. The researchers also developed an informative interface on rabies, including proper steps to take in case of exposure, to raise awareness and improve accessibility for potential users.

User Authentication Module. This module controls users' safe access to the system. Users must first establish an account, then log in with their credentials, and provide protection, limited access, and secure identity verification by ensuring that only validated, registered users can access the system.

Mapping Module. This module is a map of accessibility features of places such as clinics, hospital clinics, and private clinics. The system provided details, such as the clinic's name, location, address, and directions, to help guide the user to the destination. However, this module does not offer real-time navigation. Users needed to refresh the application to update their directions, as real-time navigation required a paid service.

Incident Report Module. This module allows the user to read an incident report for an animal bite, including the animal's name, type of animal (e.g., dog, cat), animal classification (e.g., stray, pet, wild), and the incident date.

Information Module. This module provides vital information about animal rabies. This module also helps users become more aware and prepared

in case of animal rabies exposure. It could also eliminate misconceptions about animal rabies, such as incorrect assumptions about how the disease is spread and how to treat it.

Profile Module. This module allows users to manage and update their personal information within the system. In this module, users can edit key details such as their name, contact number, and email address, and can also log out of their account.

Dashboard Module. This module provides data analytics and visualizations designed to help administrators monitor and assess reported cases within the system. It presents key information through charts, graphs, and summary statistics, enabling administrators to easily track trends, count incidents, and make informed decisions.

Incident Management Module. In this module, administrators can efficiently handle and examine user-submitted incident reports, obtain thorough information about each reported case, evaluate its contents, and take the necessary action. It also enables administrators to generate reports for additional evaluation, documentation, or record-keeping.

Clinic Management Module. This module allows administrators to manage clinic information in the system efficiently. The admins can register new clinics and view or review existing ones, ensuring that the database of healthcare facilities remains accurate and up to date, supports organized records, and helps maintain effective facility management

Notification Module. This module allows administrators to view whenever new incident reports are submitted to the system. This ensures that

admins are informed of incoming cases, allowing them to respond, review, and take action on the incident reports.

User Management Module. This module allows administrators to view and manage all registered users in the system and to access each user's profile settings to review or update their information when necessary.

Limitation of the Study

The geographic area of the system application is restricted to public clinics, hospital clinics, private clinics, and any other establishments that provide care for animal bites in Los Baños, Laguna. On the contrary, there may be situations in which private clinics are unwilling to provide the researchers with their data or information for the project. In addition, some system functionalities may impose restrictions due to the requirement to use a billing account.

CHAPTER II

THEORETICAL FRAMEWORK OF THE PROJECT

Review of Related Literature

This theoretical framework provides the basis for comprehending the concepts and principles that direct the creation and application of the system project. It offers relevant literature, studies, articles, journals, and other helpful research papers. It presents the overall literature, which help the researchers strengthen the study.

Geographic information system (GIS)

The ancient cave paintings and, later, John Snow's cholera mapping in the 18th century demonstrate the human need to investigate and record geographic information. This historical development shows an ongoing need to comprehend and represent geographic relationships. Roger Tomlinson's invention of the first GIS in 1962 was a turning point in the 20th century, paving the way for the founding of ESRI and the advancement of GIS applications in the present era. Thus, capturing and analyzing spatial data has long been a human endeavor, as reflected in the evolution of GIS (Garcia et al., 2022).

A Geographic Information System (GIS) is a system that combines persons, data, hardware, and computer software. Data associated with locations on Earth's surface can be entered, altered, analyzed, and presented through this integration. GIS makes it easier to handle geographically referenced data by interacting with technology, software, data, and people (Ali, 2020).

Geographic Information Systems (GIS) must be defined, as they involve more than just creating maps and presenting visuals. Basic mapping elements

may be available in spreadsheet programs, but they are merely a small portion of what full-function GIS can do. As a result, it is critical to view GIS as a whole system rather than just a visual aid. In the context of this de nition, the GIS differs from other information systems in several ways. Initially, the purpose of GIS was to facilitate map creation. There are numerous ways to create maps. These include creating maps from satellite data or aerial photos, digitizing paper maps, and gathering map coordinates via surveying methods or global positioning systems (GPS) (Mennecke & Crossland, 1996).

When this technology was first introduced in the Philippines in the middle of the 1970s, its main uses were for monitoring and inventory assessment of forest, mineral, and coastal zone resources, as well as for creating baseline resource information. The application of these new technologies in communications, location and navigation, and meteorology (Leonardo et al, 2007).

Since its initial adoption by the government, Geographic Information Systems (GIS) have rapidly grown in the commercial sector. The substantial financial investment in GIS software and services is reflected in market projections, such as Longley et al.'s estimate of $15–20 billion for total worldwide expenditure in 2000 and Daratech's estimate of $1.7 billion for core GIS software and services in 2003. These projections show that GIS software and services are continuing to grow in popularity across a variety of industries, including marketing, logistics, and real estate. Even while large corporations use GIS extensively, it is growing globally, particularly in developing nations. Because GIS can enhance decision-making, influence corporate strategy, and enhance transaction processes, it is significant to information systems

researchers. The industry's explosive growth demonstrates the growing importance of GIS technology for applied researchers (Pick, 2004).

Since the 1970s, Geographic Information Systems (GIS) have evolved from a small number of specialized applications into widely available computerized systems for spatial analysis. Along with the development of geographic statistics, this expansion is credited to improvements in personal computers, supporting software, and digital base map resources such as TIGER. Both basic and advanced spatial and network-based modeling are now possible due to the resulting capabilities, including georeferencing, mapping, and spatial measurements (Wieczorek & Delmerico, 2009).

Geographic Tagging

Geographic Tagging or GeoTagging is a process of adding geographic information about digital content, such as addresses, longitude and latitude location, and/or other positional data, to “metadata” tags (Salm, 2025).

Two requirements must be fulfilled to achieve high geotagging accuracy for archival natural data: the original data must be sufficiently accurate, and an efficient geotagging technique must be developed (Nowak et al., 2021).

The global positioning system (GPS) is the source of geotagging. Content can be geographically identified thanks to geotagging. As a result, the process of identifying geographic locations improves the creation of georeferenced multimedia content, georeferenced Web resources, and geographic databases. As a result, geotagging changes how consumers engage with digital information and other users. Latitude and longitude coordinates, distance, and location names are typically added to media less by system with geotagging capabilities. Both verbal and visual elements make up

the extra data. (Amaral, 2014).

Rabies

Rabies is distinct from other viral diseases in that it can infect a wide range of hosts, including all warm-blooded animals. It is known that the bulk of Asia's developing countries are the leading cause of rabies deaths (Singh et al, 2017).

In 1942 and 1943, a documented rabies outbreak occurred in the northern highlands of Carchi (Ortiz-Prado). From Carchi to Imbabura, the infectious feast spread across the country over the next ten years. Cotopaxi, Manabí, Tungurahua (1945), Chimborazo and Bolivar (1947), Los Ríos (1948), and Guayas areas (1949) followed (Bilal, 2021).

The two components of the bullet-shaped virus family Rhabdoviridae are responsible for rabies. The ribonucleocapsid core is found in the second, more functional portion, whereas the viral envelope, which is the first, is thought to be more structural. Although saliva can spread the virus through broken skin or mucous membranes, the most common way it spreads is through the bite of an infected animal, both domestic and wild. Ingestion, transplacentally, inhaling the virus in aerosolized form, and even receiving an organ donation are other ways to become infected. (Koury & Warrington, 2022). In the Philippines, where dog-mediated diseases are common, rabies is still a serious problem. The Philippines has one of the highest rates of human rabies in Asia, despite efforts to prevent and manage the disease (Smith et al, 2024).

A deadly virus called rabies is contracted by animal bites, mostly from dogs. It remains a public health concern despite vaccination efforts, particularly in the Philippines. (Cabrera, et al. 2024).

The rabies virus enters the nerves and is transported centripetally to the dorsal root ganglion via axoplasmic flow. At the site of vaccination, it produces distinctive prodromal symptoms and paresthesias and multiplies further in the dorsal root ganglion. It then travels to the brain, where it replicates rapidly. The virus then spreads to nearly every organ via different neurons, including the adrenal medulla, cornea, pancreas, hair follicle nerve twiglets, and salivary gland. This causes the virus to shed in the saliva (Dutta, 2014).

Rabies Symptoms

Hydrophobia is the most common sign of furious rabies. Seizures and episodic hyperactivity are further symptoms—hyperventilation, pituitary dysfunction, aerophobia, autonomic dysfunction, and cardiac arrhythmias. The symptoms of paralytic rabies include symmetric paralysis, acute in ammatory polyneuropathy, hypophonia, and ascending paralysis, weakness, and quadriparesis (Shadani et al., 2022).

Rabies usually takes 2 to 3 months to develop, although it can take from 1 week to 1 year, depending on the viral load and the site of virus entry. Fever, pain, and strange or inexplicable tingling, pricking, or scorching sensations at the wound site are typical early symptoms of rabies. The virus causes gradual and severe in ammation of the brain and spinal cord as it spreads throughout the central nervous system. (World Health Organization - WHO, 2024).

Human rabies is known to occur in five main stages: incubation, prodrome, acute neurologic phase, coma, and death (or, in rare cases, recovery). Once clinical signs or symptoms appear, no particular antirabies medication is helpful. The respiratory system (sore throat, cough, and dyspnea), gastrointestinal system (anorexia, dysphagia, nausea, vomiting

abdominal pain, and diarrhea), or central nervous system (headache, vertigo, anxiety, apprehension, irritability, and nervousness) may be involved in these symptoms, which are frequently nonspecific (general malaise, fever, and fatigue). A history of a recent animal bite, along with other notable abnormalities, including agitation, photophobia, priapism, heightened libido, insomnia, nightmares, and sadness, should raise the possibility of rabies. (Rupprecht. 1996).

Rabies (Post-Exposure Prophylaxis)

PEP, or post-exposure prophylaxis, is a successful strategy for preventing rabies, which can be extremely deadly for those who are exposed. The pure Vero cell rabies vaccine (PVRV) is presently being used in Malaysia. Other vaccine types and regimens are nevertheless commercially accessible. This systematic review aimed to compare the efficacy of various PEP vaccines, regimens, and administration methods among Asian populations. (Ahmad, et al. 2022).

It is the most common way of preventing human rabies. It involves immediately cleaning the wound with soap and water, monitoring the attacking animal typically a dog or cat, ten days, and administering immunobiologicals (vaccine and antirabies serum or immunoglobulin), the indication of which varies depending on the type of exposure, the wound's characteristics, the species, and the attacking animal's status (whether it be a dog, cat, or wild animal), as advised by the Ministry of Health (Estima, 2022).

Amoxicillin-clavulanate treatment is necessary for all bites to the hands or feet, bites in immunocompromised people, bites that already exhibit infection symptoms, and bites with a puncture characteristic (Maniscalco, 2022).

To minimize human rabies deaths in poor rural communities, improved vaccine access and knowledge of PEP protocols are important. Reducing the canine rabies reservoir is also essential for eliminating human rabies in populations (Preiss et al., 2018).

Rabies is completely preventable with prompt post-exposure treatment, broad dog vaccination, and community awareness, even though it is always dangerous once the symptoms appear. Launched in 2018 by prominent international groups, the global "Zero by 30" project seeks to remove dog-mediated human rabies deaths by 2030. To achieve this, focused efforts are needed in Africa and Asia, which have the highest rates of rabies-related deaths, especially in vulnerable populations (Nadal et al, 2023).

Rabies (Pre-Exposure Prophylaxis)

In both adults and children, pre-exposure rabies prophylaxis proved immunogenic and safe. According to available data, booster intervals can be prolonged up to 10 years, and shorter regimens and regimens with fewer doses are both safe and immunogenic. According to limited cost-effectiveness studies, pre-exposure prophylactic campaigns would often not be cost effective given current vaccination and delivery costs. Only Peru and the Philippines have put in place suitable national programs, even though pre-exposure prophylaxis has been recommended for high-risk groups. Accelerated regimens and new vaccines may make delivery easier and more affordable in the future. (Kessels, et al. 2016).

Pre-exposure prophylaxis (PrEP) against rabies was made available to schoolchildren as part of a Department of Health program that ran from January to February 2012. In accordance with the World Health Organization's standard

PrEP schedule, this program administered three intradermal doses of purified Verocell rabies vaccine. Trained municipal health workers gave vaccinations at schools, and the Provincial Health Office kept a record of the children who received the shots for future use (Deray et al., 2018).

The key to preventing rabies is the concept of prime (PrEP) and boost (PEP) vaccination. Investing in PrEP before a once-in-a-lifetime trip (ideally as a young adult) can have a significant impact. Subjects always need extra rabies PEP after a possible rabies exposure to boost adaptive and "trained" immunity, unlike with all vaccine-preventable illnesses. This enhanced response following PrEP and after PEP , in fact, produce higher concentrations of specific antibodies against rabies more quickly and a stronger response than after PEP alone. The primary goal is not to focus on the persistence of detectable and/or protective antibodies, but rather to rely on immune memory triggering antibody titers through anamnestic responses to additional PEP doses (Soentjens, Croughs, 2021).

Mobile Application

A small, wireless computing device that has developed software applications, unlike desktops, laptops, or computers. These apps are created specifically for a given platform, whether web-based, native, hybrid, or a third category that combines elements of both native and web apps. Nowadays, mobile apps play a vital role in today's digital age. They are an essential part of most people's daily lives, serving as social networking and entertainment, as well as for productivity and business purposes. (Katie & Wigmore, 2024).

A smartphone app called RabiApp was created to help medical professionals manage cases of animal bite `wounds by empowering them to

make well informed decisions. Alpha and beta testing of the app produced positive results, demonstrating its usability and dependability. Notably, this is the first mobile application developed in India exclusively to guide healthcare practitioners in addressing animal bite wounds. Its creation marks a substantial breakthrough in medical technology, aiming to enhance patient care and expedite clinical decision-making (Swathi et al., 2020).

The Animal Bite Mapping (ABMap) System was created to collect and evaluate information on animal bites, ensuring that all relevant factors are considered. The community is informed about biting incidents by the thorough reports that the system produces. Top-level management can access and review these reports for efficient decision-making because they are systematically maintained on the ABTC database server. Following authentication, the information can be made publicly available by integrating it into a Geographic User Interface (GUI). The system also serves as a means of communicating important information to the nation, the local community, and even the world at large (Guiquing, 2020).

In Haiti, IBCM (Integrated Bite Case Management) is a successful community based monitoring system that lowers rabies-related human mortality by offering a structure and direction for bite case investigations. Higher user acceptance, more effective data reporting and analysis, and better data quality and completeness have been achieved through Haiti's use of the REACT app. Even amid social, political, economic, and natural upheavals, rabies-endemic nations might use Haiti's eIBCM as an affordable way to reduce human rabies mortality and improve data consistency and openness. (Schrodt et al. 2023).

When the first personal digital assistants (PDAs) were released in the 1980s, mobile applications became available. However, until smartphones advanced to execute more complex programs in the twenty-first century, such applications did not advance much beyond the most fundamental and practical of functions (such as clocks and calculators). Third-generation (3G) mobile networks also enabled the download of files larger than those for text messaging and email. In the 2000s, a new business emerged when smartphone makers permitted the download of third-party mobile apps. The subsequent growth in consumer mobile app options transformed how people commute, work, play, and shop. (Adam, 2025). Mobile application software engineering places significant focus on user interface (UI) design. Making the most of the restricted screen space is difficult for developers; UI design becomes more crucial than ever. Related to User Experience (UX), developers say that varying user interaction patterns, screen layouts, and smaller displays significantly influence how mobile apps are designed, leading to lower application acceptance and usage. (Chandi, et al. 2017).

Synthesis of the Study

The reviewed literature highlights the significance of technologies such as geographic information systems (GIS), geotagging, mobile applications, and rabies prevention in community and public health. These modern technologies have been widely adopted across sectors, such as urban planning, thereby improving residents' quality of life and resource management. Geotagging is an extension of GIS that plays a crucial role in data accuracy and enhances geospatial multimedia applications, creating an interactive and informative experience useful across various fields. However, in countries like the

Philippines, rabies remains a critical public health issue, and dog-mediated transmission is also prevalent. The related literature emphasizes the importance of preventive measures against animal rabies, such as post-exposure prophylaxis (PEP) and pre-exposure prophylaxis (PrEP), which help reduce its impact. In addition, mobile applications have become instrumental in public health management, as demonstrated by systems such as RabiApp, the Animal Bite Mapping (ABMap), and the Integrated Bite Case Management (IBCM). These applications facilitate data collection, decision-making, and improved healthcare. With the integration of GIS and mobile applications, animal rabies management shows positive results in monitoring cases, improving reporting accuracy, and ensuring timely medical interventions.

Relevance of the Study

This study helped local health departments manage rabies cases in the community, especially with regard to reporting animal bites and preventing and controlling rabies. The goal of the project was to create a system that makes use of GIS, geotagging, and mobile application technologies in order to efficiently track, record, and handle rabies-related cases, including timely and accurate data reporting. The solution increased the accuracy of location-based data and the general effectiveness of case monitoring by integrating these technologies. Additionally, the introduction of mobile technology made it possible for consumers and medical professionals to access pertinent health information more easily and submit reports more quickly. These technologies greatly enhanced data collecting, decreased human error, and raised case report correctness and dependability. In addition to supporting local government units (LGUs) by offering a thorough, user-friendly, and

technologically advanced method to managing rabies cases and enhancing community health response, this study addressed current gaps in rabies monitoring and control.

Table 1. Literature Matrix

Author(s) & Year	Title	Key Findings
Garcia et al. (2022)	Geographic Information Systems and their Application in Social Sciences	GIS plays a crucial role in decision making, spatial data analysis, and geographic research. It enables mapping of disease outbreaks, urban planning, and disaster response.
Smith et al. (2024)	Rabies in the Philippines: A Call to Action	According to the report, rabies is still a serious problem in the Philippines because of low public awareness, inadequate surveillance, and inadequate vaccination. The results suggest that improving tracking systems and enhancing community education can help control the disease more effectively.
Amaral (2014)	Geotagging	Geotagging allows precise location tagging of multimedia content, improving accessibility and usability. It has applications in social media, research, and public health.
Swathi et al. (2020)	Development of a Mobile App for Rabies Prophylaxis	The app improved decision making among healthcare professionals by guiding them in rabies wound classification, post-exposure prophylaxis (PEP) recommendations, and patient monitoring.
Schrodt et al. (2023)	Electronic Application for Rabies Management	The app used in Haiti enhanced rabies surveillance, improved data reporting accuracy, and facilitated faster response times. It also increased user adoption in the community.

Conceptual Framework

Figure 1 illustrates how users interact with the GeoRabies system to report animal bites and locate vaccination clinics. The system is designed to streamline case reporting, identify high-risk areas, and improve access to

vaccination facilities. Users access the GeoRabies mobile application, which provides key functionalities such as Animal Bite Report Submission, Clinic Location Information, and Rabies and Guidelines. Furthermore, the collected data is stored in a centralized database, which integrates additional information, such as animal bite records from the LGU and clinic details (location, name, address, etc.). The system processes the data and provides key outputs, including case reports and analytics for monitoring trends. This allows health authorities to make informed decisions regarding rabies prevention and control efforts. Through this framework, the GeoRabies system enhances public health response and ensures efficient management of animal bite incidents.

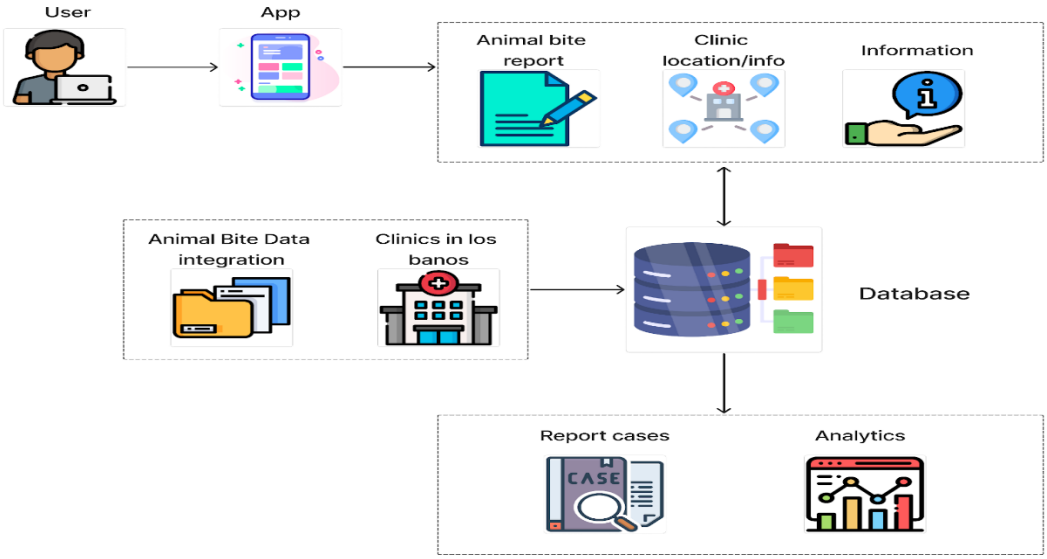


Figure 1. Conceptual Framework

CHAPTER III

METHODOLOGY

This chapter discusses the required Software, Specification of Mobile Phone, Project Locale, Project Design, Project Development, Evaluation Matrix, as well as the Georabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with Integration of GIS in Los Baños.

Project Design

System Architecture

Figure 2 illustrates the system architecture of the proposed application, comprising several interconnected modules for both users and administrators. The system is structured around key functionalities, including user authentication, incident reporting, mapping, information access, clinic management, data analytics, and user administration. To use the system, the users must first create an account through the User Authentication Module. Once authenticated, users can log in and access the features available to them based on their role.

The user side of the system comprises ve main components. The Incident Reporting Module enables users to file animal-bite incident reports. The Mapping Module allows users to view the accessibility of nearby clinics or vaccination centers and provides directional guidance to their locations. The Information Module serves as a hub for informative pages. Also, it provides guidelines for administering first aid in animal-bite situations, ensuring users are equipped with immediate steps to follow in emergencies. The Pro le Settings Module allows users to manage their personal information and log out of the system when needed.

On the Administrator side, several management modules are available to support system oversight and data analysis. The Dashboard Module provides visualizations to monitor reported cases and trends. Through the Incident Management Module, administrators can view detailed reports submitted by users and generate official reports for further review or record-keeping. The Clinic/Vaccination Center Management Module enables admins to register new clinics and view existing ones, maintaining an up-to-date database of facilities. The Notification Module is where the administrators view the latest incident reports as they are submitted. Additionally, the User Management Module allows the admin to view all registered users. Like regular users, administrators can also access their Pro le Settings Module to update their information.

All modules are connected to a centralized database that efficiently stores and retrieves data. This architecture ensures that all system components work together seamlessly, supporting both operational functionality and the project's overall goal—providing a reliable platform for reporting incidents and locating essential health services for animal bite cases.

To put it briefly, the system's Administrator side is thoughtfully designed to facilitate efficient management and well-informed strategic decision-making. It gives administrators consolidated management over clinic data, animal bite incident reports, user data, and system analytics. The platform is a vital tool for managing healthcare resources, keeping an eye on reported instances, and guaranteeing data accuracy when all of its modules work together harmoniously. Additionally, the technology facilitates prompt incident reactions and enhances clinic and local health unit cooperation. All things considered, it

is essential to the resolution of animal bite cases, the provision of essential medical services, and the advancement of the larger goal of rabies prevention and control.

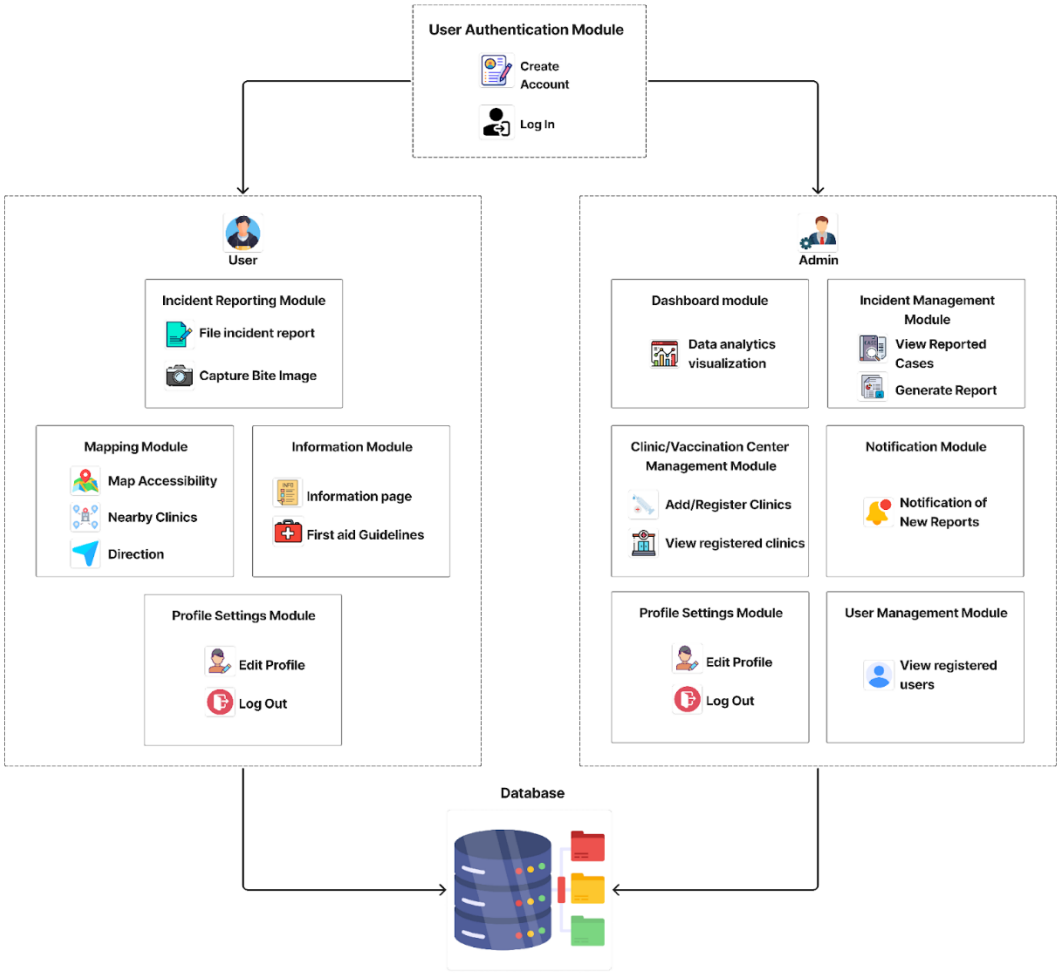


Figure 2. System Architecture

Materials

To develop and finalize the Georabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with Integration of GIS in Los Baños, the researchers would use the following materials and equipment.

Software

Listed below is the software used by the developer team to develop the mobile android-based application.

Figma

Figma was used for designing the user interface (UI) and user experience (UX) of the system. It provided a collaboration platform that allowed developers and designers to work together in real time and create wireframes and prototypes. This helped visualize the layout and navigation of mobile applications, ensuring a user-friendly and intuitive interface before actual development.

Geographic Information System

The project's Geographic Information System (GIS) was applied and integrated to enable the mobile application to access maps via OpenStreetMap. This enhances the system's ability to provide directions and guide users to specific locations. The maps include several layers, such as base maps, roads, streets, place markers, and other useful location information. Integrating these layers makes it easier for users to find nearby clinics or hospitals that provide treatment for animal bites.

Firebase Database

A cloud-based backend platform that offers scalable, real-time database services, in contrast to local databases, which make it possible for data to be safely stored, synchronized, and accessed over the internet, facilitating easy data exchange between users and devices. Without requiring manual server maintenance, it enables user authentication, real-time updates, and effective data management. Animal bite reports, user data, and other pertinent system data are stored and managed in this project using Firebase. This implementation supports real-time monitoring and enhances accessibility for the mobile Android-based system while guaranteeing data availability,

consistency, and reliability.

Visual Studio Code

Visual Studio Code, also known as "VS Code," is an open-source, free, and lightweight code editor. Visual Studio Code supports many languages, including JavaScript, Python, Java, C++, and others. Visual Studio Code blends the ease of use of a source code editor that offers robust debugging, code completion, and developer tools.

Expo Go

Expo Go is a sandbox environment designed for testing and running React Native apps on mobile devices. Without having to set up a whole development environment, it enables developers to rapidly experiment with creating native iOS and Android apps.

React Native

React Native is an open-source framework developed by Facebook for building mobile applications using JavaScript and React. It allows developers to create native-style apps for both iOS and Android platforms using a single codebase. This framework brings the best parts of React, a popular JavaScript library for building user interfaces, to mobile development.

Hardware

Table 2 shows the minimum requirements for smartphones to operate the produced application flawlessly and error-free are listed in this section to maximize accessibility for a larger range of users, the developers made sure the program is compatible with lesser versions of the Android operating system. However, Android 10 and higher is the suggested operating system for best performance, stability, and full feature support. For the program to work fully,

including real-time data submission, map services, and synchronization with the central database, a reliable internet connection is necessary. Accurate location tracking, report submission, and clinic information retrieval all depend on internet availability. All things considered, these minimal specifications guarantee constant system functionality and dependable performance for users on all compatible devices.

Table 2. Mobile Specification

Minimum Requirements	Specification
Android OS	At least Android 10 and above
RAM	8 GB
Processor	Quad - core
Internet Speed	At least 2 Mbps

Table 3 shows the minimum specifications for a computer or laptop to be used in developing the project. The processor (CPU) is an Intel i5 7th Gen or above if applicable; 8 GB of RAM for system execution; 256 GB SSD storage for faster application or software loading; and Windows 10 operating system.

Table 3. Computer Specification

Minimum Requirements	Specification
Processor(CPU)	Intel i5-7th gen. and above
RAM	8 GB Ram
OS	Windows 10
Storage	256 GB SSD

Figure 3 illustrates the Agile Scrum Methodology, a dynamic, iterative project management approach that enhances flexibility, teamwork, and ongoing

system enhancement. This approach ensures that important functionalities are implemented effectively while maintaining a user-focused design in the creation of GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with GIS Integration in Los Baños. The project uses a sequence of two to four-week sprints to methodically develop and improve the system. Every sprint focuses on delivering the features that enable iterative improvements based on testing and user input. Flexibility, effective work on management, and user needs responsiveness are guaranteed by this methodical approach.

Project Development

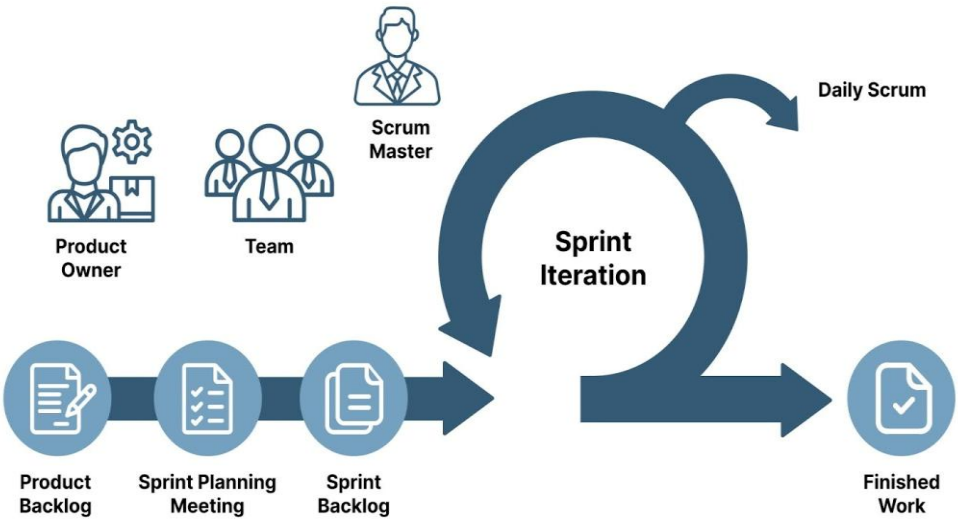


Figure 3. Agile Scrum Methodology

Product Backlog

Based on the product backlog, the researchers determined the characteristics and specifications necessary for the system's mobile component. The features that must be created for end users are listed below, along with an expected completion date.

Table 4 shows the prioritized list of tasks and system features needed to enable administrative operations in the Product Backlog for the Administrator. To help with development planning, each backlog item is arranged according to how important it is to system functionality and is accompanied by an estimated time of completion. The backlog concentrates on critical administrative functions like data analytics, incident monitoring, user administration, and secure login access.

Table 4. Product Backlog (Administrator)

PRIORITY	ITEMS	EST HR/DAY
1	As an Administrator, I want to have an admin login and dashboard.	2 Hours
2	As an Administrator, I want to manage users.	2 Hours
3	As an Administrator, I want to view animal bite report cases.	1 Hour
4	As an Administrator, I want to have Data Analytics Visualization.	2 Hours
5	As an Administrator, I want to generate reports of animal bite cases.	2 Hours
6	As an Administrator, I want to add/register clinic location and information.	2 Hours
7	As an Administrator, I want to view registered clinics.	1 Hour
8	As an Administrator, I want to have notification of new reports.	2 Hours
9	As an Administrator, I want to manage profile settings.	1 Hour

Table 5 illustrates the user product backlog. This displays the elements of the system that are meant for end users. These backlog items are ranked according to their projected development effort and significance for user interaction and functionality. Because it is necessary for user authentication and safe system access, the registration and login feature is given top importance. After logging in, individuals can report animal bite incidences by sending in

pertinent information and images, which is essential for precise documentation and prompt action. Additionally, the system enables users to view registered clinics via an integrated map, making it simple for them to find local medical institutions. The ability to get clinic directions facilitates navigation and makes it easier for users to get the right medical treatments. In addition, the system offers rabies education and emergency first-aid instructions to assist users in taking the proper and prompt action after an animal bite. Last but not least, the profile management function improves account control and security by enabling users to modify their personal data and password. In order to provide efficient animal bite reaction and prevention, the user product backlog often concentrates on accessibility, incident reporting, healthcare navigation, and user awareness.

Table 5. Product Backlog (Users)

PRIORITY	ITEMS	EST HR/DAY
1	As a User, I want to register and log in my account.	1 Hour
2	As a User, I want to report animal bite incidents by submitting details.	1 Hour
3	As a User, I want to view the clinics in the integrated map	1 Hour
4	As a User, I want to access the direction of the clinics.	1 Hour
5	As a User, I want to view emergency first aid instructions for bite cases and supporting information about rabies.	1 Hour
6	As a User, I want to edit my profile and change password.	1 Hour

Sprint Planning

Table 6 shows the sprint backlog for the administrator. The development team’s prioritized tasks for the administrator role during the sprint are detailed in the table. As stated in the sprint backlog, items are selected from the product

backlog and transferred to the sprint backlog, including features that must be delivered as specified in the sprint backlog and the functions the developers are expected to accomplish in the current sprint. This process ensures that the development team focuses on high-priority tasks. These processes are carefully planned to align with the project goals, timelines, and deliverables

Table 6. Sprint Backlog (Administrator)

PRIORITY	ITEMS	TASK	EST HR/DAY
1	As an Administrator, I want to have an admin login and dashboard.	Develop and design login page and admin dashboard.	2 Hours
2	As an Administrator, I want to manage users.	Develop and design user management page.	1 Hour
3	As an Administrator, I want to view animal bite report cases.	Develop and design animal bite report cases page.	1 Hour
4	As an Administrator, I want to have Data Analytics Visualization.	Develop and design data analytics visualization page.	2 Hours
5	As an Administrator, I want to generate reports of animal bite cases.	Develop and design generate reports of animal bite cases page.	2 Hours
6	As an Administrator, I want to add/register clinic location and information.	Develop and design add/register clinic location and information page.	1 Hour
7	As an Administrator, I want to view registered clinics.	Develop and design view registered clinics.	1 Hour
8	As an Administrator, I want to have notification of new reports.	Develop and design notification of new reports page.	2 Hours
9	As an Administrator, I want to manage profile settings.	Develop and design manage profile settings.	1 Hour

Table 7 illustrates the sprint backlog for the user; which describes the precise development tasks needed to implement the system's user-related features within a sprint. The user product backlog is the source of each backlog item, which is then divided into manageable tasks that direct the development process. The design and development tasks required to finish each user story

are shown in the "Task" column and the expected time needed to complete each work is shown in the "Est Hr/Day" column. The sprint backlog ensures that user functionalities such as account registration incident reporting, clinic map viewing, navigation assistance, access to emergency first-aid instructions, and profile management are developed in an organized and prioritized manner. This sprint backlog guarantees that crucial user features are finished within the allotted sprint timeframe and aids the development team in effectively monitoring progress.

Table 7. Sprint Backlog (User)

PRIORITY	ITEMS	TASK	EST HR/DAY
1	As a User, I want to register and log in my account.	Develop and design registration and log in page.	1 Hour
2	As a User, I want to report animal bite incidents by submitting details.	Develop and design animal bite incidents by submitting details.	1 Hour
3	As a User, I want to view the clinics in the integrated map	Develop and design the clinics for user to view in the integrated map	1 Hours
4	As a User, I want to access the direction of the clinics.	Develop and design access direction of the clinics page.	1 Hour
5	As a User, I want to view emergency first aid instructions for bite cases.	Develop and design emergency first aid instructions for bite cases page.	1 Hour
6	As a User, I want to edit my profile and change password.	Develop and design edit profile and change password page.	1 Hour

Product Owner

A product owner is responsible for the project's result. They are responsible for defining product requirements, monitoring the backlog, and ensuring the scrum team is aware of the goals. The product owner also serves

as the customer's voice, ensuring the scrum team delivers value aligned with user requirements and corporate goals.

Scrum Team

A Scrum Team is a group that works together to deliver products. It is composed of a Scrum Master, a Product Owner, and developers. The scrum master is responsible for managing the scrum process and removing obstacles; the product owner acts on behalf of the stakeholders and ensures the product backlog is clearly stated; and the developers are the team members who do the actual work to deliver the product.

Scrum Master

A scrum master is a team leader who conducts daily meetings, sprint planning, reviews, and progress reflection, and assists the team in properly sticking to scrum practices. A scrum master ensures the team has the resources it needs to achieve, enhances teamwork, helps the team in overcoming obstacles or difficulties that prevent development, and teaches the team and stakeholders about scrum and agile best practices while protecting them from distractions so they can focus on their tasks.

Sprint Iteration

The scrum team works and accomplishes a specific set of tasks. The team also plans the next task, completes it within the set time frame, and then checks and reviews the work to ensure it meets the defined goals. This process then repeats with the next sprint, building the product incrementally.

Daily Scrum

The scrum team conducts daily scrums to plan, check, and coordinate activities, providing an opportunity to inspect the progress of developing the

system and ensure the team stays on track to meet the sprint goal.

Finished Work

This refers to a backlog item or any specific task or feature that has been completed, tested, and checked during a sprint and meets the agreed-upon DoD (Definition of Done), a set of criteria that ensures a product increment is ready.

Use Case Diagram

Figure 4 illustrates the Use Case Diagram for the GeoRabies system, outlining the key functionalities available to both users and administrators. These interactions reflect the operations each user type can perform within the application.

On the user side, the system allows individuals to create an account via Account Registration and to log in to access features. Once logged in, users can access the map and report bite incidents. They can also view nearby clinics, access an information page for guidance and first-aid resources, and map the accessibility of health clinics and centers. Additionally, users can access directions to the facilities and manage their personal information through the View/Edit Account Profile feature.

For administrators, the solution provides an extensive management dashboard that facilitates effective platform monitoring and control administrators can visualize case patterns, pinpoint high-risk regions, and facilitate data-driven decision-making with integrated data analytics tools. Administrators can add or review registered clinic information, create comprehensive reports for documentation and analysis, and view and manage reported animal bite incidences using this system. Administrators can also

control system notifications to keep track of newly submitted reports and account activity. Administrators can monitor user access, preserve data integrity, and guarantee appropriate system usage with the help of user account management services. All things considered, these administrative tasks promote rabies prevention and response initiatives, increase operational effectiveness, and offer centralized management.

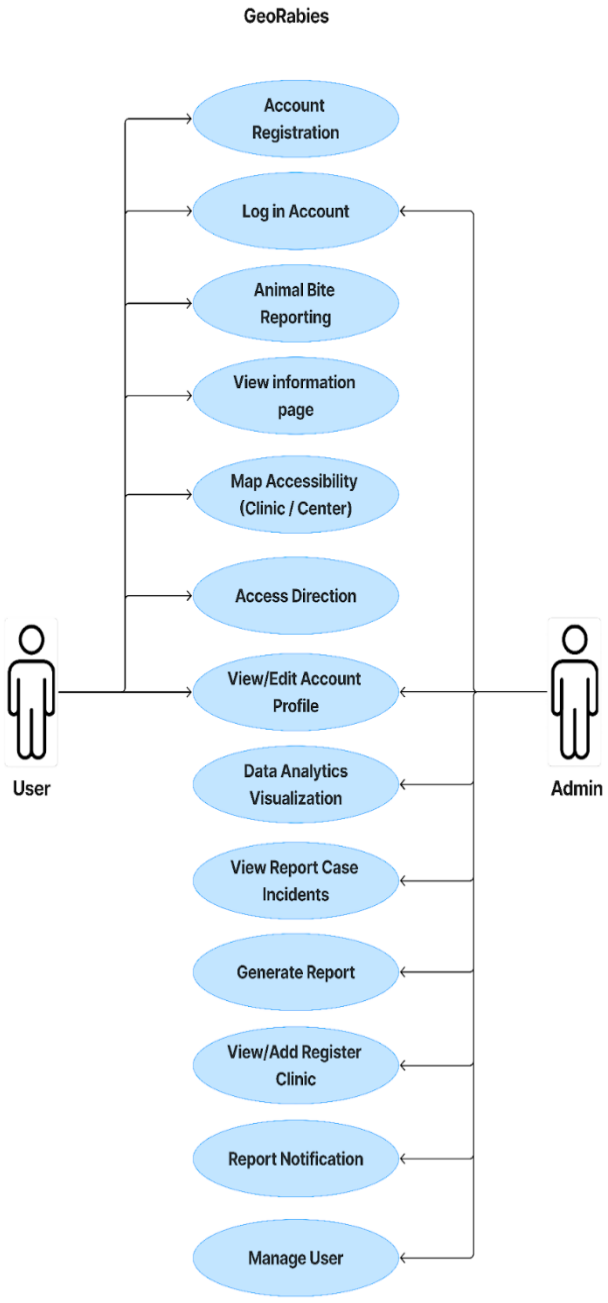


Figure 4. Use Case Diagram

System Flowchart

Figure 5 shows the user-side system flowchart, which defines the system functionalities. First, the user launches the app and navigates to the home screen, where they can use and view the map. Users can see clinics on the map and use the Nearby Clinics Button to display nearby clinics at the current user's location.

Next, in the Information Tab, users can view and read the first-aid guidelines for animal bites. Moreover, learn some facts about animal rabies. What is an animal bite? Pre-Exposure Prophylaxis, Types of animal bites, Common animals that cause bites, Animal Bite Categories, and Myth & Facts about rabies, which can enlighten users who misunderstand rabies.

Then, To add a report, The users have to log in to access and fill up the needed information such as Name, Age, Gender, Contact number, Barangay (they live in), Date (of the bite accident), Animal category (stray, pet, wild), Animal type (dog, cat, etc.), Choose bite location to point what barangay area where the user have bitten by an animal.

By entering their registered email address and password users can safely access the system and manage their accounts. After logging in the users can check their account profile, modify personal data, reset their password, and exit the program whenever necessary. Users have complete control over their personal information and account security thanks to the account management tool the users must provide basic details such name, age, gender, phone number, barangay of residence, email address, and password in order to create an account. Accurate identification and efficient communication are made possible by this information the system offers a password recovery tool that

asks users to provide their registered email address in the event that they forget their password. After that, users receive an email with a One-Time Password (OTP) that enables them to safely reset their password and get back into their account.

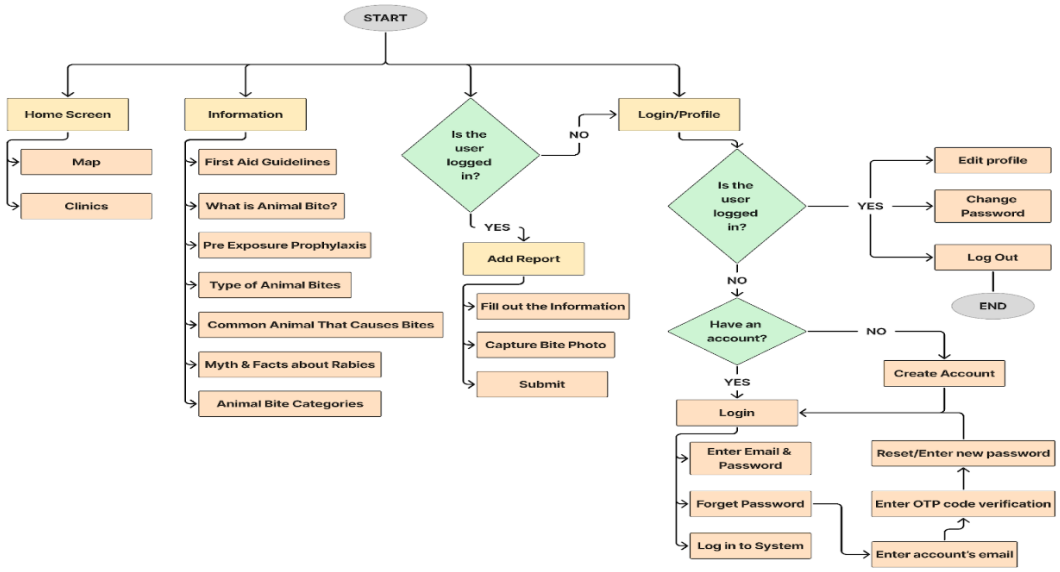


Figure 5. System Flowchart (User)

Figure 6 shows the admin-side system flowchart, which defines the system functionalities. First, the admin dashboard serves as the central interface for managing the system. Here are the data analytics for visualization, in which the researchers use different data visualizations, such as bar graphs that represent the total bite report by year and quarter, Age and Gender, and per barangay. The donut graph is also used to determine the animal category (stray, Pet, Wild) and the animal type (dog, cat, other). This visualizes the risk area for animal bites based on data and user reports. Then, Manage Incidents reports cases, from which the admin can view users' reports and generate reports. After that, the management clinic views registered clinics, including clinic information (Name, Address, Opening Schedule). Also, the Manage user allows the admin to view the user's credentials. The notification enables the

admin to be notified about updates, such as new reports and newly created user accounts. Finally, the Admin profile is where the admin can edit their name, email, and password. Once the session is done, the admin can log out, which is also located in their profile.

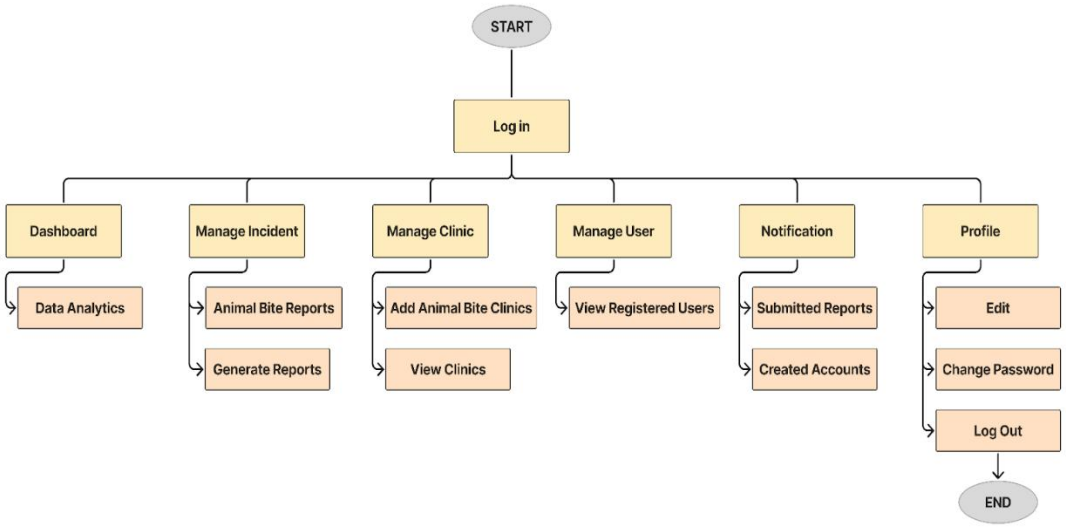


Figure 6. System Flowchart (Admin)

Project Testing and Evaluation

Software Testing

Figure 7 presents the software test plan for the project. Project testing is a critical phase in the development process that ensures the reliability, functionality, and overall quality of the system. It involves a series of structured testing activities carried out at different levels to identify and resolve defects before the system is deployed. The testing phase is divided into four major types. Unit testing, Integration testing, System testing, and Acceptance testing. Each testing type is conducted progressively to ensure each module functions properly, interacts correctly with other modules, performs as expected as a whole system, and ultimately meets the users' requirements.

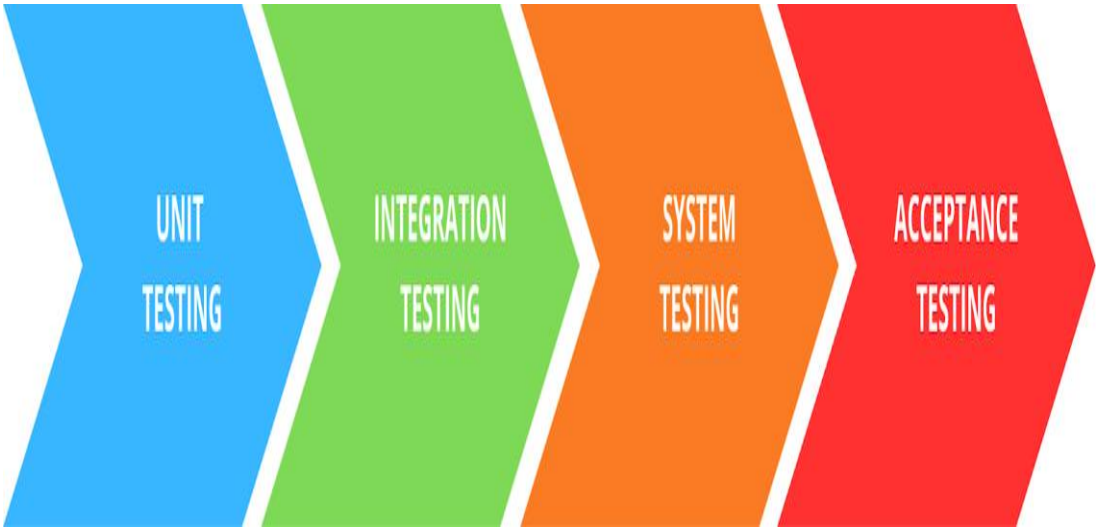


Figure 7. Software Testing

Unit Testing

Figure 8 shows the Unit Testing. The researchers conducted manual testing for each module and component, using corresponding test cases and expected outputs. This ensures that every module of the system, including the reporting form, login system, clinic locator, and vaccination record modules, functions correctly in isolation, without bugs or errors.

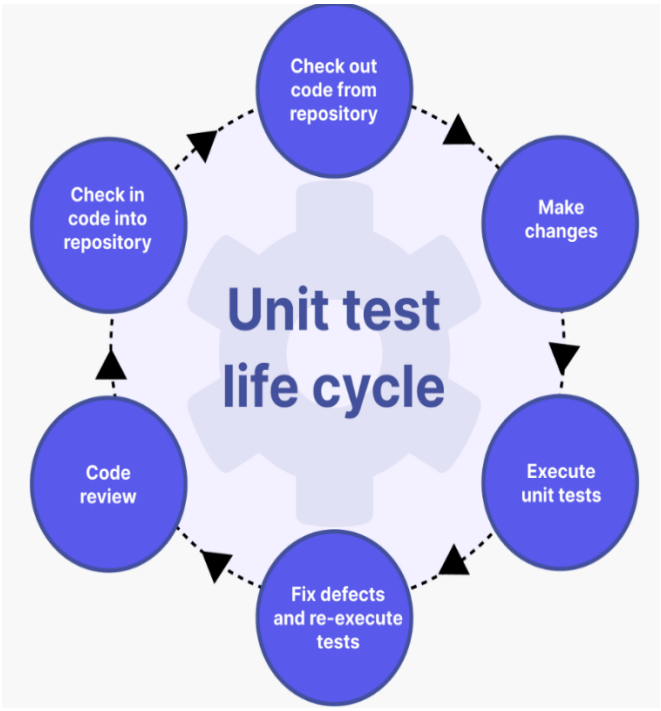


Figure 8. Unit Testing

Integration Testing

Figure 9 shows the Integration Testing. This type of testing was conducted after unit testing to check how individual modules work when combined. Modules such as user registration, report submission, map integration, and clinic database interaction were tested together to ensure they interact correctly. Integration Testing verified that there were no conflicts, data flow issues, or system logic issues.

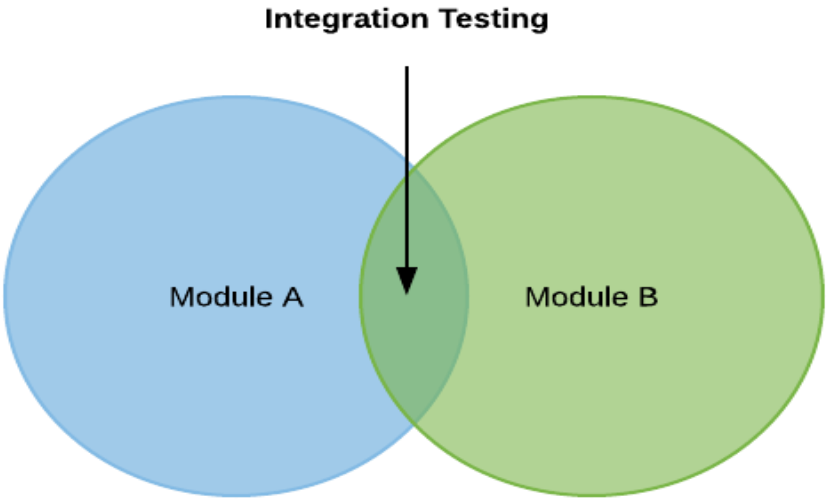


Figure 9. Integration Testing

System Testing

Figure 10 shows the System Testing, In order to guarantee system dependability, the researchers assessed the entire functionality, performance, and responsiveness of the GeoRabies mobile application during this phase. The testing confirmed that essential functions, like reporting animal bites, locating vaccination clinics, and integrating GIS-based maps, operated effectively and according to plan. To ensure precise data processing, accurate information display, and consistent performance across all usage circumstances, each functionality was evaluated. Additionally, the examination looked at the application's stability and response time under various user

interactions and during regular operation. The outcomes demonstrated that every function operated reliably and without significant mistakes or delays. All things considered, this testing step verified that the GeoRabies mobile application fulfilled its functional criteria and was prepared for user acceptance and deployment.

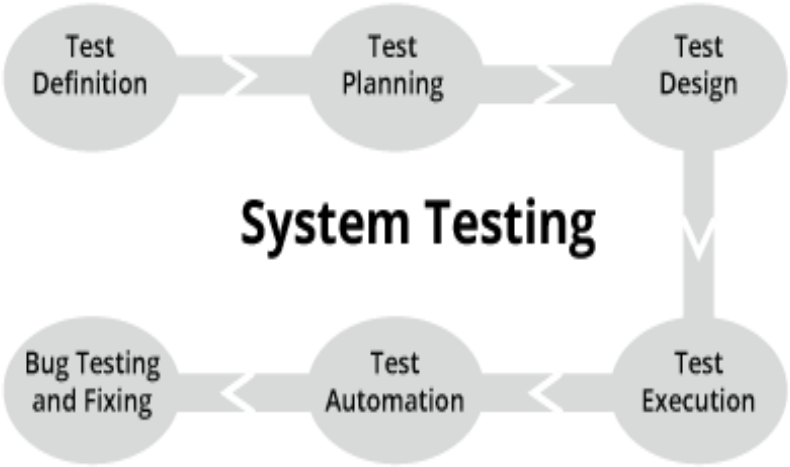


Figure 10. System Testing

Acceptance Testing

Figure 11 shows the Acceptance Testing was done before the system was deployed to make sure the application complied with all specifications and stakeholder expectations both functional and non-functional requirements, including usability, performance, and reliability, were thoroughly assessed during this phase, which functioned as the last level of verification. Key elements, such as user registration, animal bite reporting, map-based clinic location, and administrative management capabilities, were validated during the testing process. To ensure that the system performed as planned and offered a positive user experience, tester feedback was examined the acceptance testing results verified that the system was prepared for

implementation and that end users in Los Baños could use it to report animal bite incidences and find local immunization facilities.



Figure 11. Acceptance Testing

Evaluation Procedure

The researchers adopted the ISO 25010 Software Product Quality Model as the evaluation standard to assess the quality and performance of the developed system, GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with Integration of GIS in Los Baños. This model was selected to ensure that the system meets the desired quality attributes expected from a modern, user-centric mobile application.

Figure 12 shows the Software Product Quality characteristics from ISO 25010. This model provides a comprehensive framework for evaluating a software product by categorizing quality into several characteristics and sub characteristics.

Functional Suitability. The researchers evaluated whether the system performs the required functions accurately and comprehensively, including reporting animal bite incidents, locating nearby vaccination centers, and managing vaccination records.

Performance Efficiency. This aspect was assessed based on the mobile application's responsiveness and speed. The researchers checked if the

system operates smoothly on supported Android devices without lag or delay during use.

Compatibility. Since the system is a mobile application, the researchers verified that it functions correctly across various Android versions and screen sizes, ensuring it is accessible to a wide range of users in Los Baños.

Usability. Usability testing focused on how intuitive and user-friendly the app interface is. The researchers gathered feedback from users to determine if the system is easy to navigate and helps them complete tasks effectively.

Reliability. The researchers observed whether the application maintains stable performance under different usage conditions. This includes checking for app crashes or unexpected errors while users are reporting cases or browsing the map.

Security. The system’s ability to protect sensitive user information, such as location data and vaccination records, was tested. The researchers ensured that the data privacy and user confidentiality were maintained.

Maintainability. Maintainability was assessed based on how easy it is to update or modify the application if bugs or changes are needed. This ensures the system can be efficiently maintained after deployment. It also helps the developers save time and effort when performing future enhancements or xing issues, keeping the system reliable in the long run.

Flexibility. The researchers tested the application's portability by ensuring it could be installed and run smoothly across various Android environments. This makes the system more adaptable for broader use in the future.

Safety. The researchers evaluated the system’s safety by ensuring the

application does not expose users to harmful risks, especially when handling sensitive information about animal bite incidents. The system was checked to ensure that all features operate in ways that prevent misuse, support safe decision-making, and provide reliable guidance to users in health-related situations. This helps ensure that the application promotes user well-being and minimizes potential hazards during system use.



Figure 12. Evaluation Procedure

Population of the Study

The population of the study consist of residence of Los Baños, Laguna who are people that are usually prone in animal bite incident in their barangays and also IT experts who also included to the study for evaluation and test the system.

Table 8. Respondents of the Research Project

Category of Respondents	Sample
User	50
IT Experts	3
Total	55

Sampling Design

The researchers used a nonprobability sampling method, a sampling test technique, selecting residents from the surrounding area to participate in the evaluation based on factors or research criteria. The study has a total of 50 respondents and two 3 IT experts who evaluate the system application.

Data Collection Instrument

The researchers' main tool for collecting data was a formal survey, which was used to evaluate the overall efficiency, usability, and performance of the mobile Android-based system application. The purpose of the questionnaire was to gain insight into the target audience for the system, which was achieved through interviews with residents and IT specialists who tested and analyzed the mobile application.

To ensure that everyone participated in data collection process, the researchers created two different types of questionnaires: Google Forms via social media platforms for flexibility and convenience, and a hard copy survey that would be used for respondents without internet access to gather participant perspectives, experiences, and insights about the researcher's system application. The study survey questionnaire also includes Likert-scale questions. Finding trends and patterns in respondents' perspectives using the Likert scale as a research tool can help the researcher determine opinions, enabling system developers to make adjustments.

Statistical Instrument

In this study, the researchers utilized purposive sampling to gather meaningful data from selected respondents who were able to use and evaluate the GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination

Locator with Integration of GIS in the Los Baños mobile application. To assess user feedback and determine the system's usability and overall quality, a structured survey was administered as the primary data collection tool. The researchers used the Likert scale to evaluate responses, an effective technique for measuring attitudes and perceptions. The survey allowed respondents to express their level of agreement or satisfaction with system features using a four-point scale: (4) – Strongly Agree, (3) – Agree, (2) – Disagree, (1) – Strongly Disagree. This approach helped the researchers to quantify the responses and analyze the overall performance and acceptance of the system based on actual user experience. The collected data were then interpreted to ensure the accuracy, consistency, and reliability of the results in line with the research objectives.

Table 9 shows the Likert scale utilized as the survey method to evaluate the replies gathered from the participants. This scale was used to gauge how much consumers agreed or disagreed with the functionality, usability, and general performance of the system. In order to evaluate the respondents' perceptions in an organized and consistent way, each rating is associated with a particular range of composite mean values. The researchers were able to quantify user comments and assess how effectively the system fulfilled user requirements and expectations by employing a percentage-based method for response evaluation. This statistical method offered a reliable foundation for examining survey data and contrasting various assessment standards. All things considered, the application of the Likert scale guaranteed that the information acquired was trustworthy, impartial, and significant for the assessment and decision-making process of the study.

Highest Score – Lowest Score	4 – 1 = 3
Lowest Score	3 / 4 = 0.7

Table 9. Likert Scale Technique

Rating	Composition Mean	Interpretation
4	3.25 – 4.00	Strongly Agree
3	2.50 – 3.24	Agree
2	1.75 – 2.49	Disagree
1	1.0 – 1.74	Strongly Disagree

CHAPTER IV

RESULTS AND DISCUSSIONS

This chapter presents the results of the system development, testing, and evaluation conducted and discusses how the applied methods contributed to the accomplishment of its objectives.

Results by Objectives of the Study

Developed a GIS-integrated mobile application which allows users to locate nearby clinics offering bite-related treatments and access direction.

Figure 13. Shows how users may examine their present location on the map and find local clinics that provide treatment for bites and users can view important clinic information such as availability, address, price range, land line or contact information, this application also indicates the estimated time travel and service details by tapping on clinic markers on the GIS-integrated map. Additionally, the module offers a nearby clinic feature that assists users in finding the closest clinics that offer early bite related treatment, as well as a routing function that provides directions from the user's location to the chosen clinic. Users can immediately ascertain which clinics are open or able to welcome patients thanks to the inclusion of a legend indicating clinic availability. Furthermore, by offering a clear and engaging visual depiction of clinic locations in the area the map interface improves user experience particularly in an emergency, this capability helps shorten the time needed to find suitable medical facilities. All things considered, the GIS-based clinic mapping module facilitates prompt decision-making and enhances user access to critical medical services after animal bite incidences.

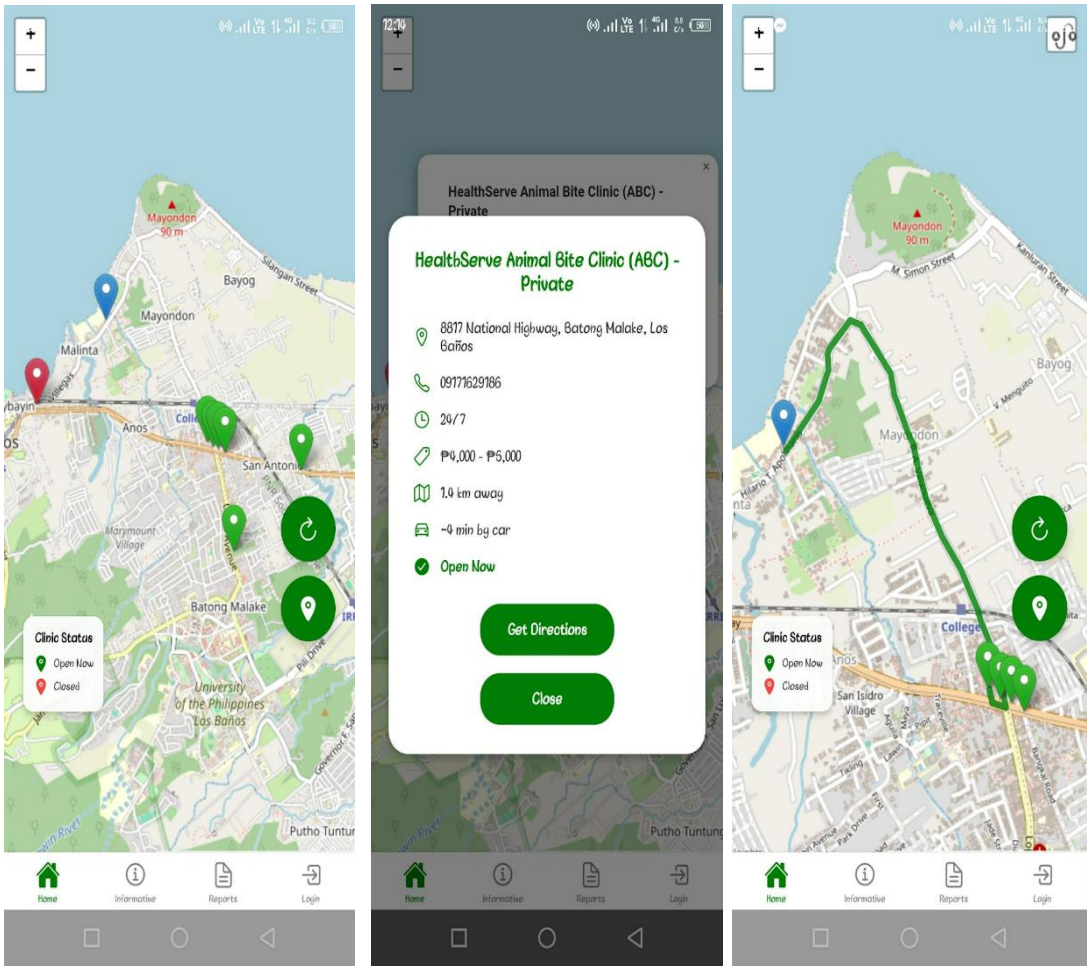


Figure 13. The Mapping Module (Homepage)

Created a centralized reporting system that enables users to report animal bite incidents.

Figure 14 shows the animal bite reporting system, which gives customers a methodical and thorough manner to submit in-depth reports of bite incidences. This tool allows users to document crucial details including the precise date of the incident, the animal's category, and the kind of animal that bit them. Additionally, the system enables users to submit more information that clarifies the incident's circumstances. The system verifies submitted data to guarantee that each report is comprehensive, educational, and consistent in order to support accuracy and appropriate assessment. Administrators can examine and keep an eye on these reports, which are safely archived. In

general, the animal bite reporting module enhances data dependability and facilitates efficient decision-making when handling and handling bite-related incidents.

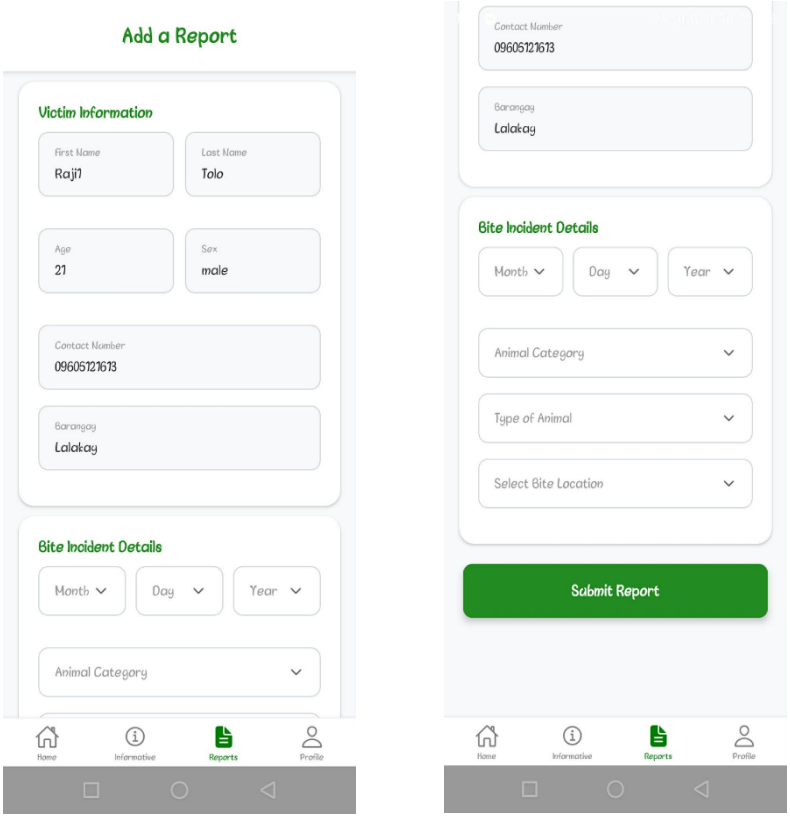


Figure 14. Animal Bite Reporting

Evaluating Result of the ISO 25010 and Technology Acceptance Model (TAM)

Table 10 presents the ISO25010 questionnaire results evaluated by IT experts. The survey yielded an overall weighted mean of 3.65 with a standard deviation of 0.29, indicating “Strongly Agree.” This suggests that the system is perceived to meet the essential software quality characteristics defined in the ISO 25010 framework to a great extent. High mean scores across most quality attributes, including functional suitability, performance efficiency, interaction capability, reliability, security, maintainability, flexibility, and safety, demonstrate that the system effectively meets critical technical requirements.

Notably, items such as Functional Suitability and Interaction Capability obtained perfect mean scores (4.00), highlighting the system's strong alignment with user expectations and its ability to support tasks efficiently and intuitively. Although Compatibility received a slightly lower mean score of 3.00 (interpreted as "Agree"), it still reflects a positive evaluation from the experts, suggesting that the system generally integrates well with other platforms and technologies, with minor areas for potential improvement. Overall, the results affirm that the system has a robust design and implementation, with reliable performance, secure operations, and strong usability. The consistently high ratings across the various ISO25010 dimensions validate the system's quality, stability, and readiness for continued deployment and use.

Table 10. Overall Result of ISO25010 Questionnaire Results (IT Experts)

ITEM	MEAN	SD	INTERPRETATION
Functional Suitability	4.00	0.00	Strongly Agree
Performance Efficiency	3.45	0.77	Strongly Agree
Compatibility	3.00	0.67	Agree
Interaction Capability	4.00	0.00	Strongly Agree
Reliability	3.67	0.29	Strongly Agree
Security	3.34	0.73	Strongly Agree
Maintainability	3.73	0.48	Strongly Agree
Flexibility	3.92	0.29	Strongly Agree
Safety	3.74	0.26	Strongly Agree
OVERALL WEIGHTED MEAN	**3.65**	**0.29**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

The results in Table 11 show that the respondents consistently thought the system was very helpful in supporting their tasks, as evidenced by the

composite mean of 3.53 in the Perceived Usefulness category, which is interpreted as "Strongly Agree." The system effectively provides the pertinent and trustworthy information required for efficient operations, as evidenced by increased accuracy in forecasting and decision-making (3.54) and increased efficiency in resource-related activities (3.56). Moreover, respondents expressed satisfaction with the system's sophisticated capabilities by giving the machine learning integration a high rating of 3.52. All things considered, the results verify that system development significantly boosts productivity and supports the features required by its target users.

Table 11. Evaluation of Perceived Usefulness by Respondents

Item	Mean	SD	Interpretation
Using the GeoRabies app will make it easier for me to report animal bite incidents.	3.56	0.54	Strongly Agree
The app will help me access direction and nearby clinics more efficiently.	3.54	0.50	Strongly Agree
The app helps me understand what to do in case of rabies exposure from being bitten by a rabid animal.	3.52	0.50	Strongly Agree
The app allows me to change or update my account details that I created.	3.50	0.51	Strongly Agree
Overall, using the app improves my effectiveness in managing rabies-related situations.	3.52	0.50	Strongly Agree
Composite	**3.53**	**0.02**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

The result in Table 12 obtained a composite mean of 3.50, also interpreted as “Strongly Agree.” This demonstrates that respondents found the system intuitive, easy to learn, and straightforward to navigate. High ratings on items such as user-friendly interface (3.52) and clear system functions (3.54)

emphasize that users experienced minimal difficulty while interacting with the platform. Additionally,the ability to complete tasks quickly (3.40) and to easily remember system functions even after some time (3.54) indicates that the interface design effectively supports both frequent and infrequent users alike. These results highlight that the application promotes accessibility and convenience, which are essential factors for improving user acceptance.

Table 12. Evaluation of Perceived Ease of Use by Respondents

Item	Mean	SD	Interpretation
Learning to use the GeoRabies app is easy for me.	3.50	0.51	Strongly Agree
Interacting with the app is clear and understandable.	3.52	0.50	Strongly Agree
Navigating through the app does not require much effort.	3.40	0.57	Strongly Agree
I find the app's features user-friendly and intuitive.	3.54	0.50	Strongly Agree
I can easily locate nearby clinics and get its direction.	3.54	0.50	Strongly Agree
Composite	**3.50**	**0.03**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74

The result in Table 13 yielded a composite mean of 3.37, interpreted as "Agree." This suggests that users generally exhibit a positive attitude toward exploring and adopting new technologies. High ratings for willingness to learn new systems (3.62) and openness to innovative platforms (3.56) reflect that most users are receptive to technological tools similar to the developed system. Although the item on experimentation with new technologies had a relatively lower mean (2.88), the overall results still indicate that respondents have a

sufficient level of technological readiness, which supports the system's potential for widespread adoption.

Table 13. Evaluation of Personal Innovation by Respondents

Item	Mean	SD	Interpretation
I am willing to use this application regularly.	2.88	0.90	Agree
I intend to use the GeoRabies app for reporting in case I got bitten by a rabid animal.	3.62	0.57	Strongly Agree
I would recommend the app to others for reporting and for them to understand more about rabies.	3.56	0.50	Strongly Agree
I am willing to continue using the app in the future.	3.40	0.61	Strongly Agree
Composite	**3.37**	**0.18**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

The result in Table 14 shows that the overall weighted mean of the TAM evaluation is 3.47, interpreted as “Strongly Agree.” This suggests that respondents’ opinions of the system's practicality, use, and individual creativity were generally favorable. The high mean score for Perceived Usefulness (3.53) indicates that users think the system does a good job of facilitating access to healthcare-related information and reporting animal bites. The Perceived Ease of Use (3.50) rating shows that even for new users, respondents judged the system to be simple to learn and use. Users’ readiness to embrace and interact with new technologies offered by the system is reflected in the Personal Innovation component, which is understood as "Agree." These results show that the designed system is likely to be adopted by its intended users, supports usability and acceptance goals, and meets user expectations. All things

considered, the TAM results confirm that the system is prepared for practical application.

Table 14. Overall Result of the TAM Survey Answered by Respondents

ITEM	MEAN	SD	INTERPRETATION
A. Perceived Usefulness	3.53	0.02	Strongly Agree
B. Perceived Ease of Use	3.50	0.03	Strongly Agree
C. Personal Innovation	3.37	0.18	Agree
OVERALL WEIGHTED MEAN	**3.47**	**0.09**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Result of Software Testing

A. Unit Testing

Table 15 shows the Mapping Module's unit tests which concentrated on confirming the dependability and functioning of every part in charge of presenting and interacting with the map. Key elements including map accessibility, legend display, user position detection, clinic marker visualization, clinic information viewing, and navigation direction generation are all being tested. Every function was tested separately to make sure it worked as planned and generated error-free, accurate results. To assess system responsiveness and performance during map-related operations, response times were also noted. The test results show that the navigation functions responded appropriately based on the user's location, the clinic data presented precisely, and the map loaded smoothly. Every test case was marked as PASSED, indicating that the Mapping Module satisfies system requirements and operates as intended. Overall, the module's stability, dependability, and readiness for integration with other system components are confirmed by the unit testing.

Table 15. Unit test of Mapping Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
Map Accessibility	The system should display the map and legend, display the location of the user and the clinics	The system successfully display the map and legend, and also display the location of the user, as well as the clinics	2s	PASSED
Get Direction	The application should be able to get the direction of the selected clinic for navigation	The application are able to get the direction of the selected clinic for navigation	1s	PASSED
Nearby Clinics	The application should indicate the nearby available clinics depends on the current location	The application successfully indicate the nearby clinics depends on the current location	1s	PASSED
Clinic Information	The application should be able to display the clinic information such as the name, location, and contacts	The application successfully able to display the clinic information such as the name, location, and contacts	1s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 16 shows the unit testing for the Information Tab, which examined the functionality of each component responsible for displaying educational content within the system. This included testing the loading of first-aid instructions, bite response guidelines, and rabies information. Each section was tested to ensure that text content displayed correctly, navigation between information categories functioned smoothly, and all resources were accessible without errors. Results showed that all elements of the Information Tab loaded properly, with no missing content or display issues encountered. Overall, the unit test confirms that the Information Tab is fully functional, stable, and capable of delivering reliable informational resources to users.

Table 16. Unit test of Information Tab Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
First Aid Guidelines	The application should display the correct information, and details	The application successfully display the correct information and details	1s	PASSED
Dropdown Information	The application should be able to show the information in the dropdown button	The application successfully able to show the information in the dropdown button	1s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 17 presents unit testing for the Animal Bite Reporting module, focused on ensuring that all input fields and reporting functions operate correctly. This included testing the submission form, where users enter the incident date, specify the animal category, select the animal type. Each input was checked for proper validation, accuracy of data capture, and error-free submission.

Table 17. Unit test of Animal Bite Reporting Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
Add Report	The system should be able to accept users input like date of incidents, animal category, type of animal, and also submission of report	The system successfully able to accept users input like date of incidents, animal category, type of animal, submission of report	1s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 18 presents unit testing for User Authentication, which examined the functionality of all components involved in user authentication. This included testing the login form fields, validating entered credentials, handling errors for incorrect inputs, and the account creation process. Each function was tested

individually to ensure that users could successfully sign in with valid credentials and receive appropriate feedback when entering invalid information.

Table 18. Unit test of User Authentication Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
User login	User should be able to login using their registered account like username and password and after validation the user should be access to the system	User successfully able to login using their registered account like username and password and after validation the user should be access to the system	1.5s	PASSED
Create Account	The user should be able to create an account by entering the input needs like name, age, gender, barangay, contact number, email, and password	The user successfully able to create an account by entering the input needs like name, age, gender, barangay, contact number, email, and password	1.3s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 19 presents the unit testing results for the Profile Settings module, This concentrated on confirming the elements in charge of handling user account data. This involved testing important features like upgrading account credentials, modifying personal information, and safely exiting the system. To guarantee correct validation, correctness, and error-free processing, every interactive component—such as input fields, save buttons, and logout options—was tested separately. According to the test findings, users were able to successfully update their data, and the system database correctly reflected all of the changes. Response times showed that logout and profile updates were handled quickly. Additionally, the logout feature operated reliably, guaranteeing that user sessions were appropriately ended for security reasons.

Overall, the Profile Settings module showed dependable performance and safe account management features, passing every unit test.

Table 19. Unit test of Profile Setting Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
Edit Profile	The user should be able about to edit or update their credential.	The user successfully able to edit or update their credential.	1.3s	PASSED
Log out	The user should be able to logout once the are finished or done using the application.	The user successfully able to logout once the are finished or done using the application.	1s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 20 presents unit tests for the Admin Dashboard, focusing on the accuracy and responsiveness of all components that display system analytics and report summaries. This included testing the loading of charts, data visualizations, and statistical summaries based on user-submitted animal bite reports. Each dashboard element was tested individually to verify that it retrieved the correct data and displayed information without errors. Overall, the Admin Dashboard passed the unit tests and proved stable, reliable, and effective at presenting system insights to administrators.

Table 20. Unit test of Admin Dashboard Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
Data Analytics Visualization	The system should display data analytics visualization of users. Like table and graphs	The system successfully displays data analytics visualization of users. Like table and graphs	1s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 21 presents unit testing for the Manage Incident module, concentrating on making sure administrators are able to appropriately examine and handle reports of animal bites received by users. To verify accuracy, functionality, and system responsiveness, each essential feature—such as displaying animal bite report cases and producing reports for record-keeping—was evaluated separately. The system operates effectively when loading reports and creating documents, according to the documented response times. In order to make sure that the compiled data appropriately mirrored the submitted report cases without errors or data loss, the report creation feature was specifically assessed. According to the test results, every tested scenario generated the anticipated results and was designated as PASSED, signifying successful implementation. All things considered, the Manage Incident module showed dependable performance, effective data processing, and deployment readiness for managing incident records.

Table 21. Unit test of Manage Incident Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
Animal Bite Report Cases	The system should be able to display the reports of the user, and review their report	The system successfully able to display the reports of the user, and review their report	1.3	PASSED
Generate Report	The system should generate report base on the report case and for record-keeping.	The system successfully generate report base on the report case and for record-keeping.	1.4	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 22 Shows unit testing for the Manage Clinics module, focused on verifying the functionality of all features used by the administrator to manage clinic information. This included testing the process of registering new clinics,

inputting details such as clinic name, address, operating hours, availability, price range, and geographic coordinates, and ensuring that each clinic was accurately displayed on the map for users. Each function was tested individually to confirm correct data entry, storage, and retrieval, as well as proper integration with the mapping module. The results showed that all clinic management features operated smoothly, with data updates reflecting immediately on the map and no errors occurring during registration or editing. Overall, the Manage Clinics module passed the unit tests, confirming it is reliable, accurate, and fully functional.

Table 22. Unit test of Manage Clinic Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
View Registered Clinic	The system should display the registered clinic information after inputting all the information needs.	The system successfully displayed the registered clinic information after inputting all the information needs.	1s	PASSED
Add Clinic	The system should be able to register and add new clinic by inputting the clinic information.	The system successfully able to register and add new clinic by inputting the clinic information.	1.2s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 23 shows unit testing for the Manage Users module, which evaluated each function responsible for displaying and managing user accounts within the system. This included testing the retrieval and display of user information, verifying that the list of registered users loaded correctly, and ensuring that administrators could view account details without errors. Overall, the Manage Users module passed the unit tests, demonstrating stability and reliability in managing user records.

Table 23. Unit test of Manage User Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
View Registered user	The system should display the registered users who created an account and their information.	The system successfully displayed the registered users who created an account and their information.	1s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Table 24 Shows unit testing for the Notification module focused on ensuring that system-generated notifications functioned correctly. This included testing the display of the new user account and newly submitted animal bite reports. Each notification was checked for accuracy, proper timing, and correct linking to the relevant module or record. Results showed that all notification features worked as expected. Overall, the Notification module passed the unit tests and is confirmed to be functional.

Table 24. Unit test of Notification Module

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
Notification	The application should be able to display the newly reports and newly created accounts	The application successfully able to display the newly reports and newly created accounts	1.3s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

B. Integration Testing

Table 25 presents the Integration Testing phase, which aims to validate the proper interaction and data flow among all interconnected modules within the GeoRabies Application. Unlike unit testing, which focuses on individual components, integration testing ensures that the combined modules work seamlessly as a complete system. During this phase, the development team

executed test procedures to verify communication among essential system features, including the user authentication module, incident reporting module, mapping module, information module, profile setting module, dashboard module, incident management module, clinic management module, notification module, and user management module. These interactions were carefully examined to ensure that data is transmitted, stored, and displayed accurately. Critical integration points, such as user authentication enabling correct dashboard access, submitted incident reports reflecting in the admin's Manage Incident module, clinic locations properly syncing with the mapping interface, and notifications triggered by user or admin actions, were all tested. The system was checked for consistency, responsiveness, and correct handling of both valid and invalid data inputs across modules.

Table 25. Integration Testing Result

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
User Authentication	The system should have a secured registration when creating account and login for users as well to access to the system	The system successfully implements a secured registration and login. The system has validation on empty inputs and wrong inputs	1s	PASSED
Animal bite reporting module	The system should have the correct input fields for user to fill up their incident report details.	The system successfully have the correct input fields for user to fill up their incident report details.	1.3s	PASSED
Mapping Module	The system should display the map and the user can able to see clinic location and their current location. As well they can locate nearby clinics and get its direction.	The system successfully display the map and the user can able to see clinic location and their current location. As well they can locate nearby clinics and get its direction.	1.5s	PASSED
Information Module	The system should display the correct	The system successfully display the correct	1.2s	PASSED

	information about first aid and guidelines about rabies, as well as the user can read and see the drop down information about rabies related.	information about first aid and guidelines about rabies, as well as the user can read and see the drop down information about rabies related.		
Profile Setting Module	The system should be able to let the user to edit their credentials or update their email and password, as well they can log out after using the application.	The system successfully able to let the user to edit their credentials or update their email and password, as well they can log out after using the application.	1.1s	PASSED
Dashboard Module	The system should be able to display the data analytics module, like the table graph, pie graph, and the data are based on the user incidents reports	The system successfully able to display the data analytics module, like the table graph, pie graph, and the data are based on the user incidents reports	1.3s	PASSED
Incident Management Module	The system should be able to view all the report from the user and can generate report	The system successfully able to view all the report from the user and can generate report	1.2s	PASSED
Clinic Management Module	The system should display the clinics that are registered and can register new clinic by registering the clinic information, as well it can update or edit.	The system successfully display the clinics that are registered and can register new clinic by registering the clinic information, as well it can update or edit.	1.3s	PASSED
User Management Module	The system should display all registered user and able to see the details of their accounts	The system successfully display all registered user and able to see the details of their accounts	1.1s	PASSED
Notification Module	The system should be able to show the newly created account and newly report from the user and can be filter by barangay.	The system successfully able to show the newly created account and newly report from the user and can be filter by barangay.	1.4s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

C. System Testing

Table 26 shows the system testing results, Activities were carried out to evaluate the entire functionality, dependability, and performance of the GeoRabies mobile application as a fully integrated system. This testing step made sure that all system modules functioned together and fulfilled the requirements set during the planning and design stages after individual unit and integration tests were finished the development team assessed the application's end-to-end workflow from both user and administrator viewpoints in order to confirm real-world usability. Verifying accurate user registration and login redirection, making sure dashboards exhibited up-to-date and trustworthy data, and confirming that animal bite reports were correctly filed, stored, and reflected in the administrative data analytics were all part of the testing procedure.

Additionally, the system was verified to ensure that clinic data updated by administrators was appropriately shown on the integrated map with the right locations and details. Notification features were also assessed to guarantee that newly established accounts and reported occurrences were promptly and accurately presented.

To guarantee stability and appropriate error handling, system behavior was evaluated in a variety of scenarios, including typical usage, edge cases, wrong inputs, and unexpected user activities. To gauge the effectiveness of the system during crucial activities, response times were noted. All things considered, the system testing phase confirmed that the GeoRabies mobile application functions as planned, provides precise data, is stable, and is ready for deployment.

Table 26. System Testing Result

TESTED CASE	EXPECTED OUTPUT	ACTUAL OUTPUT	RESPONSE TIME(s)	REMARKS
User registration and login	The user should be redirected login screen after creating a account. The user should redirect to the map screen after log in, and the admin should be redirected to the admin dashboard	The user successfully redirected to the login screen after creating an account and the user are abled to redirect to the map screen after log in, as well the admin successfully redirected to the admin dashboard	1.5s	PASSED
Animal Bite Report	The report should be display in the admin incident management and reflect to the admin dashboard as data analytics visualization	The report successfully displayed in the admin incident management and reflect to the admin dashboard as data analytics visualization	1s	PASSED
Edit profile	The user or admin should be able to update or edit only their own credential	The user or admin successfully abled to update or edit only their own credential	1.4s	PASSED
Generate Report	Generating report should be reflects on the user's reports and to the data analytics.	Generating report successfully be reflected on the user's reports and to the data analytics.	1.3s	PASSED
Add Clinics	Adding clinics to the clinic management should be reflect to the map and show its information and location.	Adding clinics to the clinic management are successfully reflected to the map and show its information and location.	1.2s	PASSED
Notification	Upcoming report and created account should be display in the notification	Upcoming report and created account are successfully be displayed in the notification	1.3s	PASSED
User Management	Created account should be display in this module and able to edit and delete	Created account successfully displayed in this module and able to edit and delete	1.3s	PASSED

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

CHAPTER V

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

This chapter summarizes the study, including the findings, and presents the conclusion of the Georabies Application system, highlighting the overall performance and effectiveness of the developed system. This provides a conclusion that reflects on the objectives. This chapter also offers recommendations for future researchers to improve and enhance the system's application.

Summary

The study developed the GeoRabies Application system, a mobile Android platform designed to assist potential users in reporting animal bite incidents and to provide access to essential guidelines, first aid, and educational information on rabies-related issues. It also integrates a geographic information system (GIS) and a mapping module using Leaflet to provide geolocation, geotagging, Directions, and the ability to locate nearby clinics. The system application is also designed to manage animal bite reports and monitor the behavior of animal bite incidents using data analytics and visualization, enabling the generation of reports for record-keeping and use for decision making. The researchers managed to conduct a system evaluation using the ISO25010 software quality standard in terms of functional suitability, performance efficiency, usability, reliability, security, and other quality characteristics, including the Technology Acceptance Model, which assessed the key quality and demonstrated the function effectiveness, ease of use, and the relevance when it comes to rabies prevention and control.

Conclusions

Based on the results of the study, the researchers successfully achieved the objectives of the project:

1. Developed a GIS-integrated mobile application which allows users to locate nearby clinics offering bite-related treatments.

The integration of GIS into the mobile application was implemented using OpenStreetMap, enabling users to locate nearby clinics that treat animal bites and access navigation directions.

2. Created a centralized reporting system that enables users to report animal bite incidents and access direction.

The researchers created a centralized reporting system that allows users to report animal bite cases and access directions to nearby clinics for immediate treatment. This feature also serves as a practical tool for both users and administrators in managing animal bite incidents.

3. Evaluated the developed system using ISO 25010 and conduct a user acceptance evaluation based on the technology Acceptance Model (TAM)

The ISO/IEC 25010 software quality evaluation model, which addresses important quality attributes such functional appropriateness, performance efficiency, compatibility, dependability, security, maintainability, adaptability, and safety, was used to evaluate the system application. This assessment made that the system functioned well under various operating conditions and complied with globally accepted software quality standards. To find possible problems and verify system stability, every quality aspect was thoroughly examined. The Technology Acceptance Model (TAM) assessment was carried out to gauge user acceptance in terms of perceived utility, perceived usability,

and individual creativity. The GeoRabies Application System is practical, dependable, and user-friendly, according to the findings of both assessments. Overall, these results support the system's suitability for real-world deployment by confirming that it satisfies defined quality requirements and closely matches user expectations.

Recommendations

Based on the results and findings of the study, the researchers propose the following recommendations for future researchers who wish to enhance, expand, or further develop the GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with Integration of GIS in Los Baños:

1. **Integration of Google Account Sign-Up and Authentication**

   Future researchers are encouraged to integrate Google account sign-up and login to simplify user registration and improve accessibility. This enhancement would provide users with a more convenient authentication method and a more efficient password recovery option, thereby improving the overall usability of the system.

2. **Development of Multi-Platform Compatibility**

   Since the current version of the GeoRabies application operates exclusively on Android devices, future researchers may consider developing versions compatible with other operating systems and device types. Expanding Compatibility—such as creating versions for iOS and other platforms—would allow a wider range of users to access and benefit from the system’s functionalities.

3. **Implementation of Offline Route Navigation Features**

   Although the application supports offline maps, its route navigation (“Get

Directions”) feature requires an active internet connection. Future researchers are encouraged to explore enabling full offline navigation so that users can still obtain directions to clinics even without internet access. This would be especially beneficial in remote or low-connectivity areas where timely access to navigation is critical.

4. **Enhancing Real-Time Navigation in Maps**

   Currently, the GeoRabies application provides navigation via OpenStreetMap, allowing users to view routes to vaccination clinics. However, real-time tracking of the user's location is hampered because the map only updates their location if the page is refreshed. True real-time navigation should be implemented so the user's location constantly updates as they walk or move, similar to how it works in Google Maps. This will enhance the user experience by providing more accurate guidance for travel to clinics, especially in emergencies, and make the system more reliable for mobile users on the go.

5. **Development of a Web-Based Administrative System**

   Future researchers may consider creating a web-based administrative dashboard to complement or replace the current mobile-admin setup. Administrators typically manage operations more efficiently using desktop or browser-based environments; therefore, developing a web interface would enhance usability, streamline data management, and support better administrative control.

6. **Establishment of a Regular System Maintenance Framework**

   To maintain long-term system reliability and relevance, future researchers are advised to implement a structured and regular

maintenance framework. Monthly system checks—covering updates, performance optimization, bug fixes, and security enhancements—would ensure that the application remains functional, responsive, and aligned with evolving technological requirements.

REFERENCES

Asif Bilal, (2017) Rabies is a Zoonotic Disease: A Literature Review https://www.researchgate.net/publication/358802264_Rabies_is_a_Zoonotic_Disease_A_Literature_Review

Ahmad N, Nawi AM, Jamhari MN, Nurumal SR, Mansor J, Zamzuri M'IA, Yin TL, Hassan MR. (2022) Post-Exposure Prophylactic Vaccination against Rabies: A Systematic Review. Iran J Public Health. 967-977. doi: 10.18502/ijph.v51i5.9412. PMID: 36407746; PMCID: PMC9643235. https://pmc.ncbi.nlm.nih.gov/articles/PMC9643235/

B. B. Mennecke, M. D. Crossland, (1996) Geographic Information Systems: Applications and Research Opportunities for Information Systems Researchers. https://www.researchgate.net/publication/3624069_Geographic_information_systems_applications_and_research_opportunities_for_information_systems_researchers

Chandi, L., Silva, C., Martínez, D., & Gualotuña, T. (2017). Mobile application development process: A practical experience. Paper presented at the 12th Iberian Conference on Information Systems and Technologies (CISTI), Lisbon, Portugal. https://doi.org/10.23919/CISTI.2017.7975825

Deray, R., Gripon, S., Ulanday, C., Roces, M. C., Amparo, A. C., Attlan, M., Demont, C., Kieffer, A., Miranda, M. E. G., Rivera, C. (2018). Protecting children from rabies with education and pre-exposure prophylaxis: A school-based campaign in El Nido, Palawan, Philippines https://www.researchgate.net/publication/322219384_Protecting_children_from_rabies_with_education_and_pre-exposure_prophylaxis_A_school-based_campaign_in_El_Nido_Palawan_Philippines

Ershad Ali, (2020) Geographic Information System (GIS): Definition, Development, Applications & Components https://www.researchgate.net/publication/340182760_Geographic_Information_System_GIS_Definition_Development_Applications_Components

Estima NM, Wada MY, Rocha SM, Moraes DS, Ohara PM, Vargas A, Assis DM. (2022) Description of human anti-rabies post-exposure prophylaxis care notifications in Brazil, 2014-2019. Epidemiol Serv Saude. e2021627. doi: 10.1590/S2237-96222022000200002. PMID: 35730814; PMCID: PMC9887991. https://pmc.ncbi.nlm.nih.gov/articles/PMC9887991/#sec9

G.R. Cabrera, M.C. S. Guillermo, S. Evangelio , E.O. Diamante, (2024) A Centralized Mobile App for Dog Rabies Vaccination Reporting by Private

and City Veterinarians in Davao City: Issue Joint Conference & Symposium on Health
https://ojs.upmin.edu.ph/index.php/jcsh/article/view/70

Guiquing, F. L. (2021). Animal bite mapping (ABMAP) using mobile apps. Psychology and Education Journal, 58(2), 3398–3407. https://psychologyandeducation.net/pae/index.php/pae/article/view/3398

Hanna, K. T., & Wigmore, I. (n.d.). Mobile app. TechTarget. Retrieved from https://www.techtarget.com/whatis/definition/mobile-app

I. Amaral, (2014) In book: Encyclopedia of Social Media and Politics (pp.574-575) Chapter: Geotagging Publisher: Sage Publications Editors: Kerric Harvey
https://www.researchgate.net/publication/270887898_Geotagging

Kessels JA, Recuenco S, Navarro-Vela AM, Deray R, Vigilato M, Ertl H, Durrheim D, Rees H, Nel LH, Abela-Ridder B, Briggs D. (2016) Pre-exposure rabies prophylaxis: a systematic review. Bull World Health Organ. 2017 Mar 1;95(3):210-219C. doi: 10.2471/BLT.16.173039. PMID: 28250534; PMCID: PMC5328107. https://pmc.ncbi.nlm.nih.gov/articles/PMC5328107/

Koury R, Warrington SJ. (2022) Rabies In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK448076/

L. Leonardo, BA. Crisostomo, JA. Solon, P. T. Rivera, A. Marcelo, J. Villasper, (2007) Geographical information systems in health research and services delivery in the Philippines Geospatial Health 1(2):147-55 DOI:10.4081/gh.2007.263 SourcePubMed LicenseCC BY-NC 4.0 https://www.researchgate.net/publication/23155117_Geographical_information_systems_in_health_research_and_services_delivery_in_the_Philippines

M.M. Nowak, K. Słupecka, B. Jackowiak, (2021) Geotagging of natural history collections for reuse in environmental research, Ecological Indicators, Volume 131,108131,ISSN 1470-160X, https://doi.org/10.1016/j.ecolind.2021.108131.

Maniscalco K, Edens MA. (2022) Animal Bites. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK430852/

Nadal, D., Bote, K., Masthi, R., Narayana, A., Ross, Y., Wallace, R., & Abela, B. (2023). Rabies post-exposure prophylaxis delivery to ensure treatment efficacy and increase compliance. IJID one health, 1, 100006. https://doi.org/10.1016/j.ijidoh.2023.100006

Preiss, S., Chanthavanich, P., Chen, L. H., Marano, C., Buchy, P., van Hoorn, R., … Mukherjee, P. (2018). Post-exposure prophylaxis (PEP) for rabies with purified chick embryo cell vaccine: a systematic literature review and meta-analysis. Expert Review of Vaccines, 17(6), 525–545. https://doi.org/10.1080/14760584.2018.1473765

Pick, J. (2004). Geographic information systems: A tutorial and introduction. Redlands. Retrieved from https://www.academia.edu/69074884/Geographic_information_systems_A_tutorial_and_introduction

Patrick Soentjens, Mieke Croughs, (2021) Simplified rabies pre-exposure prophylaxis in last-minute travellers, Journal of Travel Medicine, Volume 28, Issue 1, taaa185, https://doi.org/10.1093/jtm/taaa185

Radicelli-García, C., Floril, M., Cevallos, P., & Bodero Poveda, E. (2022). Geographic information systems and their application in the social sciences: A literature review. Journal of Geography and Cartography, 5(1). https://doi.org/10.24294/jgc.v5i1.1679

Rupprecht CE. (1996) Rhabdoviruses: Rabies Virus. In: Baron S, editor. Medical Microbiology. 4th edition. Galveston (TX): University of Texas Medical Branch at Galveston; Chapter 61. Available from: https://www.ncbi.nlm.nih.gov/books/NBK8618/

Salm, M. (2025). Geo-tagging. Better Evaluation. Retrieved from https://www.betterevaluation.org/methods-approaches/methods/geo-tagging

Singh, R., Singh, K. P., Cherian, S., Saminathan, M., Kapoor, S., Manjunatha Reddy, G. B., … Dhama, K. (2017). Rabies – epidemiology, pathogenesis, public health concerns and advances in diagnosis and control: a comprehensive review. Veterinary Quarterly, 37(1), 212–251. https://doi.org/10.1080/01652176.2017.1343516

Smith, C., Ortal, A., Girasol, M., Suzuki, S., & Coughlan, C. (2024). Rabies in the Philippines: A call to action. The Lancet Regional Health – Western Pacific, 49, 101156. https://www.thelancet.com/journals/lanwpc/article/PIIS2666-6065(24)00150-0/fulltext

Shadani S, Ghasemi F, Kamali M, Navaeifar MR, Alikhani A, Alian S (2022) Rabies Virus Infection: A Case Series and Literature Review. Journal of Pediatrics Review 10(4):315-320. http://dx.doi.org/10.32598/jpr.10.4.369.5

Swathi, M., Sabeena, S., Abdulmajeed, J., Pattanaik, S., Dsa, O., Shubha, H. S., . . . Arunkumar, G. (2020). Development and evaluation of a mobile app for guiding rabies prophylaxis among health-care professionals in

India. Indian Journal of Community Medicine, 45(4), 473. https://doi.org/10.4103/ijcm.ijcm_510_19

Schrodt, C. A., Dilius, P., Gibson, A. D., Crowdis, K., Fénelon, N., Ross, Y., Bonaparte, S., Joseph, H. C., & Wallace, R. M. (2023). Electronic application for rabies management improves surveillance, data quality, and investigator experience in Haiti. Frontiers in veterinary science, 10, 1052349. https://doi.org/10.3389/fvets.2023.1052349

T. Dutta, (2014) Rabies: An overview International Journal of Advanced Medical and Health Research 1(2):39 DOI:10.4103/2349-4220.147998 LicenseCC BY-NC-SA 4.0 https://www.researchgate.net/publication/293801831_Rabies_An_overview

Volle, A. (2025, January 23). app. Encyclopedia Britannica. Retrieved from https://www.britannica.com/technology/mobile-app

Wieczorek WF, Delmerico AM. (2009) Geographic Information Systems. Comput Stat. 1(2):167-186. doi: 10.1002/wics.21. PMID: 20717487; PMCID: PMC2921721. https://pmc.ncbi.nlm.nih.gov/articles/PMC2921721/#abstract1

World Health Organization, Rabies (2024) https://www.who.int/news-room/fact-sheets/detail/rabies =

APPENDIX A

(Survey Questionnaire)

Technology Acceptance Model (TAM) Questionnaire

Georabies: Mobile Android-Based Animal Bite Reporting and

Vaccination Locator with Integration of GIS

RESEARCHERS: Chester Jay Abas, Ragimar Tolo, Vince Leo Poblete

Instruction: *Please answer each statement by checking the box that best represents your level of agreement.*

CATEGORY I: Perceived Usefulness (PU)	**Strongly Agree**	**Agree**	**Disagree**	**Strongly Disagree**
Questions	**4**	**3**	**2**	**1**
Using the GeoRabies app will make it easier for me to report animal bite incidents. *(Makakatulong sa akin ang paggamit ng GeoRabies app upang mas madaling maulat ang mga insidente ng kagat ng hayop.)*	☐	☐	☐	☐
The app will help me access direction and nearby clinics more efficiently. *(Makakatulong ang app sa akin na mas mabilis mahanap ang direksyon at malalapit na klinika.)*	☐	☐	☐	☐
The app helps me understand what to do in case of rabies exposure from being bitten by a rabid animal. *(Tinutulungan ako ng app na maintindihan kung ano ang dapat gawin sakaling ma-expose sa rabies dahil sa pagkagat ng isang hayop na may rabies.)*	☐	☐	☐	☐
The app allows me to change or update my account details that I created. *(Pinapahintulutan ako ng app na baguhin o i-update ang aking mga detalye sa account na aking ginawa.)*	☐	☐	☐	☐
Overall, using the app improves my effectiveness in managing rabies-related situations. *(Sa pangkalahatan, pinapabuti ng paggamit ng app ang aking pagiging epektibo sa pamamahala ng mga sitwasyong may kinalaman sa rabies.)*	☐	☐	☐	☐

CATEGORY II: Perceived Ease of Use (PEOU)	Strongly Agree	Agree	Disagree	Strongly Disagree
Questions	**4**	**3**	**2**	**1**
Learning to use the GeoRabies app is easy for me. *(Madali para sa akin ang matutunan ang paggamit ng GeoRabies app.)*	☐	☐	☐	☐
Interacting with the app is clear and understandable. *(Malinaw at naiintindihan ang pakikipag-interact ko sa app.)*	☐	☐	☐	☐
Navigating through the app does not require much effort. *(Hindi nangangailangan ng maraming effort ang pag-navigate sa app.)*	☐	☐	☐	☐
I find the app's features user-friendly and intuitive. *(Natagpuan kong user-friendly at intuitive ang mga tampok ng app.)*	☐	☐	☐	☐
I can easily locate nearby clinics and get its direction. *(Madali kong mahanap ang mga malalapit na klinika at makuha ang kanilang direksyon.)*	☐	☐	☐	☐

CATEGORY III: Behavioral Intention to Use (BI)	Strongly Agree	Agree	Disagree	Strongly Disagree
Questions	**4**	**3**	**2**	**1**
I am willing to use this application regularly. *(Handa akong gamitin ang aplikasyon na ito nang regular.)*	☐	☐	☐	☐
I intend to use the GeoRabies app for reporting in case I got bitten by a rabid animal. *(Balak kong gamitin ang GeoRabies app para mag-ulat sakaling ako'y makagat ng hayop na may rabies.)*	☐	☐	☐	☐
I would recommend the app to others for reporting and for them to understand more about rabies. *(Ire-rekomenda ko ang app sa iba para sa pag-uulat at upang mas maintindihan nila ang tungkol sa rabies.)*	☐	☐	☐	☐
I am willing to continue using the app in the future. *(Handa akong patuloy na gamitin ang app sa hinaharap.)*	☐	☐	☐	☐

APPENDIX B

(IT Specialist Survey Questionnaire)

ISO 25010 QUESTIONNAIRE

We are fourth-year Bachelor of Science in Information Technology students majoring in Web and Mobile Application Development. As part of our academic requirements, we have created an **Georabies: Mobile Android-Based Animal Bite Reporting** and **Vaccination Locator with Integration of GIS** for the local community, following the ISO/IEC 25010 Software Quality Model.

To determine the system's performance and overall quality, we are conducting an evaluation focusing on key criteria such as usability, functionality, security, and user experience.

We sincerely appreciate your participation in this assessment. Please rate each item according to your experience with the system using the 4-point scale provided. All responses will be kept strictly confidential and utilized solely for academic purposes.

Thank you very much for your time and cooperation.

RESEARCHERS: Chester Jay Abas, Ragimar Tolo, Vince Leo Poblete

Direction: *Check the rubric box that corresponds to your evaluation.*

Category 1: Functional Suitability	**Strongly Agree**	**Agree**	**Disagree**	**Strongly Disagree**
Sub Category	**4**	**3**	**2**	**1**
Functional Completeness: 1. The system provides all necessary functions to manage and optimize resources.	☐	☐	☐	☐
Functional Correctness: 2. The system produces accurate and consistent outputs based on input data.	☐	☐	☐	☐
Functional Appropriateness: 3. The system's functions are relevant and helpful in achieving operational goals.	☐	☐	☐	☐

Category 2: Performance Efficiency	**Strongly Agree**	**Agree**	**Disagree**	**Strongly Disagree**
Sub Category	**4**	**3**	**2**	**1**
Time Behavior: 4. The system responds quickly to user requests and actions.	☐	☐	☐	☐
Resource Utilization: 5. The system efficiently uses available hardware and software resources.	☐	☐	☐	☐
Capacity:	☐	☐	☐	☐

6. The system maintains good performance even with multiple users or large data sets.

Category 3: Compatibility	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Co-Existence:				
7. The system can run smoothly alongside other software without conflict	☐	☐	☐	☐
Interoperability:				
8. The system can effectively exchange and use data with other applications or platforms.	☐	☐	☐	☐

Category 4: Interaction Capability	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Appropriateness Recognizability:				
9. The interface clearly shows how functions should be used.	☐	☐	☐	☐
Learnability:				
10. New users can easily learn to use the system.	☐	☐	☐	☐
Operability:				
11. The system is easy to operate and navigate.	☐	☐	☐	☐
User Error Protection:				
12. The system prevents or corrects common user mistakes.	☐	☐	☐	☐
User Engagement:				
13. The system design motivates users to continue using it.	☐	☐	☐	☐
Inclusivity:				
14. The system is accessible to all types of users.	☐	☐	☐	☐
User Assistance:				
15. The system provides adequate help, guides, or documentation.	☐	☐	☐	☐
Self-Descriptiveness:				
16. The system clearly describes its features and operations.	☐	☐	☐	☐

Category 5: Reliability	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Faultlessness: 17. The system performs tasks without unexpected errors.	☐	☐	☐	☐
Availability: 18. The system is accessible and operational when needed	☐	☐	☐	☐
Fault Tolerance: 19. The system continues to work properly even if minor errors occur.	☐	☐	☐	☐
Recoverability: 20. The system can recover quickly after a failure.	☐	☐	☐	☐

Category 6: Security	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Confidentiality: 21. The system keeps sensitive information protected from unauthorized users.	☐	☐	☐	☐
Integrity: 22. The system prevents unauthorized modification or loss of data.	☐	☐	☐	☐
Non -Repudiation: 23. The system ensures that user actions cannot be denied or altered.	☐	☐	☐	☐
Accountability: 24. The system tracks user actions accurately for accountability.	☐	☐	☐	☐
Authenticity: 25. The system verifies the identity of users properly.	☐	☐	☐	☐
Resistance: 26. The system resists unauthorized access and attacks.	☐	☐	☐	☐

Category 7: Maintainability	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Modularity: 27. The system is designed in components that can be modified independently	☐	☐	☐	☐
Reusability: 28. The system's components can be reused in other applications	☐	☐	☐	☐
Analysability: 29. The system is easy to analyze to identify errors or improvement areas.	☐	☐	☐	☐
Modifiability: 30. The system can be updated or changed without affecting other components.	☐	☐	☐	☐
Testability: 31. The system can be efficiently tested after modification or updates.	☐	☐	☐	☐

Category 8: Flexibility	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Adaptability: 32. The system can adapt to changes in requirements or environments.	☐	☐	☐	☐
Scalability: 33. The system can scale to accommodate more users or data	☐	☐	☐	☐
Installability: 34. The system is easy to install and configure	☐	☐	☐	☐
Replaceability: 35. The system allows replacement or integration of components with minimal issues.	☐	☐	☐	☐

Category 9: Safety	Strongly Agree	Agree	Disagree	Strongly Disagree
Sub Category	**4**	**3**	**2**	**1**
Operational Constraint: 36. The system performs safely under defined conditions.	☐	☐	☐	☐
Risk Identification: 37. The system identifies potential risks or errors effectively.	☐	☐	☐	☐
Fail Safe: 38. The system includes fail-safe mechanisms to prevent data loss or damage.	☐	☐	☐	☐
Hazard Warnings: 39. The system provides clear hazard or error warnings when necessary.	☐	☐	☐	☐
Safe Integration: 40. The system ensures safe integration with other software or hardware.	☐	☐	☐	☐

Validation of expert:

_____ _____

Signature over printed name Date

_____

Designation/Position

_____

Office

APPENDIX C

(Result of Post Evaluation)

Evaluation of Perceived Usefulness by Respondents (50)

Item	Mean	SD	Interpretation
I am willing to use this application regularly.	2.88	0.90	Strongly Agree
I intend to use the GeoRabies app for reporting in case I got bitten by a rabid animal.	3.62	0.57	Strongly Agree
I would recommend the app to others for reporting and for them to understand more about rabies.	3.56	0.50	Strongly Agree
I am willing to continue using the app in the future.	3.40	0.61	Strongly Agree
Composite	**3.37**	**0.18**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

Overall Result of the TAM Survey Answered by Respondents

ITEM	MEAN	SD	INTERPRETATION
A. Perceived Usefulness	3.53	0.02	Strongly Agree
B. Perceived Ease of Use	3.50	0.03	Strongly Agree
C. Personal Innovation	3.37	0.18	Agree
OVERALL WEIGHTED MEAN	**3.47**	**0.09**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.25 – 4.00), three (3) Agree (2.50 – 3.24), two (2) Disagree (1.75 – 2.49), and one (1) Strongly Disagree (1.0 – 1.74)

APPENDIX D

(Result of ISO 25010)

Overall Result of ISO25010 Questionnaire Results (IT Experts)

ITEM	MEAN	SD	INTERPRETATION
Functional Suitability	4.00	0.00	Strongly Agree
Performance Efficiency	3.45	0.77	Strongly Agree
Compatibility	3.00	0.67	Agree
Interaction Capability	4.00	0.00	Strongly Agree
Reliability	3.67	0.29	Strongly Agree
Security	3.34	0.73	Strongly Agree
Maintainability	3.73	0.48	Strongly Agree
Flexibility	3.92	0.29	Strongly Agree
Safety	3.74	0.26	Strongly Agree
OVERALL WEIGHTED MEAN	**3.65**	**0.29**	**Strongly Agree**

Legend: highest to lowest as follows: four (4) Strongly Agree (3.26 – 4.00), three (3) Agree (2.51 – 3.25), two (2) Disagree (1.76 – 2.50), and one (1) Strongly Disagree (1.0 – 1.75)

APPENDIX E

(Code Snippets)

```
// screens/Home.tsx
import React, { useState, useRef, useEffect, useContext } from 'react';
import {
  View,
  Text,
  StyleSheet,
  Dimensions,
  TouchableOpacity,
  Modal,
  ScrollView,
  Alert,
  ActivityIndicator,
} from 'react-native';
import { WebView } from 'react-native-webview';
import { Ionicons } from '@expo/vector-icons';
import * as Location from 'expo-location';
import { collection, onSnapshot, query } from 'firebase/firestore';
import { db } from '../src/firebase';
import { AuthContext } from '../App';
import { useRoute } from '@react-navigation/native';

const { width, height } = Dimensions.get('window');

interface Clinic {
  id: string;
  name: string;
  latitude: number;
  longitude: number;
  address: string;
  contactNumber: string;
  operatingHours: string;
  priceRange: string;
  distance: number;
  eta: number;
  isOpenNow: boolean;
}

const Home: React.FC = () => {
  const { user } = useContext(AuthContext);
  const route = useRoute<any>();

  const [selectedClinic, setSelectedClinic] = useState<Clinic | null>(null);
  const [modalVisible, setModalVisible] = useState(false);
  const [allClinics, setAllClinics] = useState<Clinic[]>([]);
  const [focusNearby, setFocusNearby] = useState(false);
  const [loading, setLoading] = useState(true);
  const webViewRef = useRef<WebView>(null);
  const [userLocation, setUserLocation] = useState<{ latitude: number;
longitude: number } | null>(null);
```

```
// Handle auto-routing from Report screen
  const autoRouteClinic = route.params?.autoShowRouteToClinic;

  const isClinicOpenNow = (operatingHours: string): boolean => {
    const now = new Date();
    const currentHour = now.getHours() + now.getMinutes() / 60;
    const dayOfWeek = now.getDay();

    if (operatingHours.includes('24/7')) return true;
    if (operatingHours.includes('Tuesday and Friday')) {
      if (dayOfWeek !== 2 && dayOfWeek !== 5) return false;
      return currentHour >= 13;
    }

    const timeMatch = operatingHours.match(/(\d+:\d\d)\s*(am|pm)?\s*[–-
]\s*(\d+:\d\d)\s*(am|pm)/i);
    if (!timeMatch) return false;

    const parseTime = (timeStr: string, period: string | undefined): number => {
      const [h, m] = timeStr.split(':').map(Number);
      let hours = h;
      if (period) {
        if (period.toLowerCase() === 'pm' && h !== 12) hours += 12;
        if (period.toLowerCase() === 'am' && h === 12) hours = 0;
      }
      return hours + m / 60;
    };

    const openTime = parseTime(timeMatch[1], timeMatch[2]);
    const closeTime = parseTime(timeMatch[3], timeMatch[4]);

    if (openTime >= closeTime) {
      return currentHour >= openTime || currentHour < closeTime;
    }
    return currentHour >= openTime && currentHour < closeTime;
  };

  const calculateDistance = (
    userLoc: { latitude: number; longitude: number },
    clinic: { latitude: number; longitude: number }
  ): number => {
    const toRad = (v: number) => (v * Math.PI) / 180;
    const R = 6371;
    const dLat = toRad(clinic.latitude - userLoc.latitude);
    const dLon = toRad(clinic.longitude - userLoc.longitude);
    const a =
      Math.sin(dLat / 2) ** 2 +
      Math.cos(toRad(userLoc.latitude)) *
        Math.cos(toRad(clinic.latitude)) *
        Math.sin(dLon / 2) ** 2;
```

```
    const c = 2 * Math.atan2(Math.sqrt(a), Math.sqrt(1 - a));
    return R * c;
  };

  const getUserLocation = async () => {
    try {
      let { status } = await Location.requestForegroundPermissionsAsync();
      let loc = { latitude: 14.178, longitude: 121.243 };

      if (status === 'granted') {
        const location = await Location.getCurrentPositionAsync({});
        loc = {
          latitude: location.coords.latitude,
          longitude: location.coords.longitude,
        };
      } else {
        Alert.alert('Location Permission Denied', 'Using default location: Los
Baños area.');
      }

      setUserLocation(loc);
      return loc;
    } catch (error) {
      Alert.alert('Error', 'Failed to get location. Using default.');
      const defaultLoc = { latitude: 14.178, longitude: 121.243 };
      setUserLocation(defaultLoc);
      return defaultLoc;
    }
  };

  useEffect(() => {
    const q = query(collection(db, 'clinics'));
    const unsubscribe = onSnapshot(q, (snapshot) => {
      const clinics: Clinic[] = snapshot.docs.map((doc) => {
        const data = doc.data();
        return {
          id: doc.id,
          name: data.name || 'Unknown Clinic',
          address: data.address || 'No address',
          contactNumber: data.contactNumber || 'N/A',
          operatingHours: data.operatingHours || 'N/A',
          priceRange: data.priceRange || 'N/A',
          latitude: Number(data.latitude) || 0,
          longitude: Number(data.longitude) || 0,
          distance: 0,
          eta: 0,
          isOpenNow: false,
        };
      });
```

```
      setAllClinics(clinics);
      setLoading(false);
    }, (error) => {
      console.error('Firestore clinics load error:', error);
      Alert.alert('Error', 'Failed to load clinics from server.');
      setLoading(false);
    });

    return () => unsubscribe();
  }, []);

  useEffect(() => {
    if (allClinics.length > 0) {
      getUserLocation().then((loc) => {
        const updatedClinics = allClinics.map((clinic) => ({
          ...clinic,
          distance: calculateDistance(loc, clinic),
          eta: Math.round(calculateDistance(loc, clinic) * 3),
          isOpenNow: isClinicOpenNow(clinic.operatingHours),
        }));

        const sorted = updatedClinics.sort((a, b) => a.distance - b.distance);
        setAllClinics(sorted);
        updateMapMarkers(loc, sorted, focusNearby);
      });
    }
  }, [allClinics.length > 0]);

  // Handle auto-routing when coming from Report screen
  useEffect(() => {
    if (autoRouteClinic && userLocation && webViewRef.current) {
      const { clinic } = autoRouteClinic;

      // Draw the route automatically
      webViewRef.current.injectJavaScript(`
        if (window.currentRoute) map.removeControl(window.currentRoute);
        window.currentRoute = L.Routing.control({
          waypoints: [
            L.latLng(${userLocation.latitude}, ${userLocation.longitude}),
            L.latLng(${clinic.latitude}, ${clinic.longitude})
          ],
          router: L.Routing.osrmv1({ serviceUrl: 'https://router.project-
osrm.org/route/v1' }),
          lineOptions: { styles: [{ color: '#007E05', weight: 6, opacity: 0.8 }] },
          createMarker: () => null,
          addWaypoints: false,
          routeWhileDragging: false,
          fitSelectedRoutes: true,
          show: false
        }).addTo(map);
```

```
        true;
      `);

      // Auto-select and open modal for this clinic
      const foundClinic = allClinics.find(c => c.id === clinic.id);
      if (foundClinic) {
        setSelectedClinic(foundClinic);
        setModalVisible(true);
      }
    }
  }, [autoRouteClinic, userLocation, allClinics]);

  const toggleNearbyFocus = () => {
    const newState = !focusNearby;
    setFocusNearby(newState);
    if (userLocation && allClinics.length > 0) {
      updateMapMarkers(userLocation, allClinics, newState);
    }
  };

  const updateMapMarkers = (
    userLoc: { latitude: number; longitude: number },
    clinics: Clinic[],
    nearbyFocused: boolean
  ) => {
    if (!webViewRef.current || clinics.length === 0) return;

    const js = `
      map.eachLayer(l => {
        if (l instanceof L.Marker && l.getPopup &&
!l.getPopup().getContent().includes('Your Location')) {
          map.removeLayer(l);
        }
      });

      L.marker([${userLoc.latitude}, ${userLoc.longitude}], {
        icon: L.icon({
          iconUrl: 'https://unpkg.com/leaflet@1.9.4/dist/images/marker-icon.png',
          iconSize: [25, 41],
          iconAnchor: [12, 41],
        })
      }).addTo(map).bindPopup('Your Location');

      var clinics = ${JSON.stringify(
        clinics.map((c) => ({
          ...c,
          distance: c.distance.toFixed(1),
          statusText: c.isOpenNow ? 'Open Now' : 'Closed',
          statusColor: c.isOpenNow ? '#007E05' : '#FF4444',
        }))
```

```
    )};

    clinics.forEach(clinic => {
      const iconUrl = clinic.isOpenNow
        ?                    'https://raw.githubusercontent.com/pointhi/leaflet-color-
markers/master/img/marker-icon-green.png'
        :                    'https://raw.githubusercontent.com/pointhi/leaflet-color-
markers/master/img/marker-icon-red.png';

      L.marker([clinic.latitude, clinic.longitude], {
        icon: L.icon({
          iconUrl,
          shadowUrl:         'https://unpkg.com/leaflet@1.9.4/dist/images/marker-
shadow.png',
          iconSize: [25, 41],
          iconAnchor: [12, 41],
          popupAnchor: [1, -34],
        })
      })
      .bindPopup(\`
        <div  style="width:240px;  padding:10px;  font-family:sans-serif;  font-
size:13px;">
          <b style="font-size:15px;">\${clinic.name}</b>
          <hr style="margin:8px 0; border:0; border-top:1px solid #eee;">
          <b>Address:</b> \${clinic.address}<br>
          <b>Contact:</b> \${clinic.contactNumber}<br>
          <b>Hours:</b> \${clinic.operatingHours}<br>
          <b>Price:</b> \${clinic.priceRange}<br>
          <b>Distance:</b>  <span  style="color:#007E05;">\${clinic.distance}
km</span>
          |  <b>ETA:</b>  <span  style="color:#007E05;">\${clinic.eta}
min</span><br>
          <b>Status:</b>  <span  style="color:\${clinic.statusColor};  font-
weight:bold;">\${clinic.statusText}</span>
        </div>
      \`)
      .addTo(map)
      .on('click', () => {
        window.ReactNativeWebView.postMessage(JSON.stringify({ id: clinic.id
}));
      });
    });

    if (clinics.length > 0) {
      const group = new L.featureGroup(clinics.map(c => L.marker([c.latitude,
c.longitude])));
      map.fitBounds(group.getBounds().pad(${nearbyFocused ? '0.2' : '0.5'}));
    } else {
      map.setView([${userLoc.latitude}, ${userLoc.longitude}], 13);
    }
```

```
      true;
    `;

    webViewRef.current.injectJavaScript(js);
  };

  const handleMarkerPress = (clinic: Clinic) => {
    setSelectedClinic(clinic);
    setModalVisible(true);
    webViewRef.current?.injectJavaScript(`
      if (window.currentRoute) { map.removeControl(window.currentRoute);
window.currentRoute = null; }
      true;
    `);
  };

  const handleGoDirection = (clinic: Clinic) => {
    if (!userLocation) return;

    webViewRef.current?.injectJavaScript(`
      if (window.currentRoute) map.removeControl(window.currentRoute);
      window.currentRoute = L.Routing.control({
        waypoints: [
          L.latLng(${userLocation.latitude}, ${userLocation.longitude}),
          L.latLng(${clinic.latitude}, ${clinic.longitude})
        ],
        router: L.Routing.osrmv1({ serviceUrl: 'https://router.project-
osrm.org/route/v1' }),
        lineOptions: { styles: [{ color: '#007E05', weight: 6, opacity: 0.8 }] },
        createMarker: () => null,
        addWaypoints: false,
        routeWhileDragging: false,
        fitSelectedRoutes: true,
        show: false
      }).addTo(map);
      true;
    `);

    setModalVisible(false);
  };

  const handleWebViewMessage = (event: any) => {
    try {
      const { id } = JSON.parse(event.nativeEvent.data);
      const clinic = allClinics.find((c) => c.id === id);
      if (clinic) handleMarkerPress(clinic);
    } catch (e) {}
  };

  const mapHtml = `
```

```
    <!DOCTYPE html>
    <html>
      <head>
        <meta name="viewport" content="width=device-width, initial-scale=1.0">
        <link rel="stylesheet"
href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css" />
        <link rel="stylesheet" href="https://unpkg.com/leaflet-routing-
machine@latest/dist/leaflet-routing-machine.css" />
        <style>
          body, html, #map { margin:0; padding:0; height:${height}px;
width:${width}px; }
        </style>
      </head>
      <body>
        <div id="map"></div>
        <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
        <script src="https://unpkg.com/leaflet-routing-machine@latest/dist/leaflet-
routing-machine.js"></script>
        <script>
          var map = L.map('map').setView([14.178, 121.243], 14);
          L.tileLayer('https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png', {
            attribution: '&copy; <a
href="https://openstreetmap.org">OpenStreetMap</a> contributors'
          }).addTo(map);
        </script>
      </body>
    </html>
  `;

  if (loading) {
    return (
      <View style={[styles.container, { justifyContent: 'center', alignItems: 'center'
}]}>
        <ActivityIndicator size="large" color="#007E05" />
        <Text style={{ marginTop: 10, color: '#333' }}>Loading clinics...</Text>
      </View>
    );
  }

  return (
    <View style={styles.container}>
      <WebView
        ref={webViewRef}
        source={{ html: mapHtml }}
        style={styles.map}
        onMessage={handleWebViewMessage}
        javaScriptEnabled={true}
        domStorageEnabled={true}
        onLoad={() => userLocation && allClinics.length > 0 &&
updateMapMarkers(userLocation, allClinics, focusNearby)}
```

```
      />

      <View style={styles.fabContainer}>
        <TouchableOpacity
          style={styles.fab}
          onPress={() => getUserLocation().then((loc) => updateMapMarkers(loc,
allClinics, focusNearby))}
        >
          <Ionicons name="refresh" size={24} color="white" />
        </TouchableOpacity>
        <TouchableOpacity
          style={[styles.fab,          styles.fabNearby,          focusNearby          &&
styles.fabNearbyActive]}
          onPress={toggleNearbyFocus}
        >
          <Ionicons name="location" size={24} color="white" />
          {focusNearby && <View style={styles.fabBadge} />}
        </TouchableOpacity>
      </View>

      <Modal
        animationType="slide"
        transparent
        visible={modalVisible}
        onRequestClose={() => setModalVisible(false)}
      >
        <View style={styles.centeredView}>
          <View style={styles.modalView}>
            {selectedClinic && (
              <>
                <ScrollView showsVerticalScrollIndicator={false}>
                  <Text style={styles.modalTitle}>{selectedClinic.name}</Text>
                  <View style={styles.infoRow}>
                    <Ionicons name="location-outline" size={18} color="#007E05" />
                    <Text style={styles.modalText}>{selectedClinic.address}</Text>
                  </View>
                  <View style={styles.infoRow}>
                    <Ionicons name="call-outline" size={18} color="#007E05" />
                    <Text
style={styles.modalText}>{selectedClinic.contactNumber}</Text>
                  </View>
                  <View style={styles.infoRow}>
                    <Ionicons name="time-outline" size={18} color="#007E05" />
                    <Text
style={styles.modalText}>{selectedClinic.operatingHours}</Text>
                  </View>
                  <View style={styles.infoRow}>
                    <Ionicons name="pricetag-outline" size={18} color="#007E05" />
                    <Text
style={styles.modalText}>{selectedClinic.priceRange}</Text>
```

```
              </View>
              <View style={styles.infoRow}>
                <Ionicons name="map-outline" size={18} color="#007E05" />
                <Text
style={styles.modalText}>{selectedClinic.distance.toFixed(1)} km away</Text>
              </View>
              <View style={styles.infoRow}>
                <Ionicons name="car-outline" size={18} color="#007E05" />
                <Text    style={styles.modalText}>~{selectedClinic.eta}    min    by
car</Text>
              </View>
              <View style={styles.infoRow}>
                <Ionicons
                  name={selectedClinic.isOpenNow  ?  'checkmark-circle'  :  'close-
circle'}
                  size={18}
                  color={selectedClinic.isOpenNow ? '#007E05' : '#FF4444'}
                />
                <Text
                  style={[
                    styles.modalText,
                    {  color:  selectedClinic.isOpenNow  ?  '#007E05'  :  '#FF4444',
fontWeight: 'bold' },
                  ]}
                >
                  {selectedClinic.isOpenNow ? 'Open Now' : 'Closed'}
                </Text>
              </View>
            </ScrollView>

            <TouchableOpacity
              style={styles.goDirectionButton}
              onPress={() => handleGoDirection(selectedClinic)}
            >
              <Text style={styles.goDirectionButtonText}>Get Directions</Text>
            </TouchableOpacity>
          </>
        )}

        <TouchableOpacity
          style={styles.closeButton}
          onPress={() => setModalVisible(false)}
        >
          <Text style={styles.textStyle}>Close</Text>
        </TouchableOpacity>
      </View>
    </View>
  </Modal>

  <View style={styles.legendContainer}>
```

```
        <Text style={styles.legendTitle}>Clinic Status</Text>
        <View style={styles.legendItem}>
          <Ionicons name="location" size={16} color="#007E05" />
          <Text style={styles.legendText}>Open Now</Text>
        </View>
        <View style={styles.legendItem}>
          <Ionicons name="location" size={16} color="#FF4444" />
          <Text style={styles.legendText}>Closed</Text>
        </View>
      </View>
    </View>
  );
};

const styles = StyleSheet.create({
  container: { flex: 1, backgroundColor: '#fff' },
  map: { width, height },
  centeredView: {
    flex: 1,
    justifyContent: 'center',
    alignItems: 'center',
    backgroundColor: 'rgba(0,0,0,0.5)',
  },
  modalView: {
    margin: 20,
    backgroundColor: 'white',
    borderRadius: 20,
    padding: 25,
    width: '90%',
    maxHeight: '85%',
    elevation: 5,
  },
  modalTitle: {
    fontSize: 20,
    fontWeight: 'bold',
    color: '#007E05',
    textAlign: 'center',
    marginBottom: 15,
  },
  infoRow: { flexDirection: 'row', alignItems: 'center', marginBottom: 12 },
  modalText: { marginLeft: 10, fontSize: 15, color: '#333', flex: 1 },
  closeButton: {
    backgroundColor: '#007E05',
    padding: 12,
    borderRadius: 20,
    marginTop: 15,
    alignSelf: 'center',
    width: '50%',
  },
  goDirectionButton: {
```

```
    backgroundColor: '#007E05',
    padding: 12,
    borderRadius: 20,
    marginTop: 10,
    alignSelf: 'center',
    width: '50%',
  },
  goDirectionButtonText: {
    color: '#fff',
    fontWeight: 'bold',
    textAlign: 'center',
    fontSize: 16,
  },
  textStyle: { color: '#fff', fontWeight: 'bold', textAlign: 'center', fontSize: 16 },
  legendContainer: {
    position: 'absolute',
    bottom: 120,
    left: 20,
    backgroundColor: 'rgba(255,255,255,0.95)',
    padding: 10,
    borderRadius: 10,
    elevation: 4,
  },
  legendTitle: { fontWeight: 'bold', fontSize: 13, marginBottom: 5, textAlign:
'center' },
  legendItem: { flexDirection: 'row', alignItems: 'center', marginBottom: 3 },
  legendText: { fontSize: 11, marginLeft: 5, color: '#333' },
  fabContainer: {
    position: 'absolute',
    bottom: 180,
    right: 20,
    gap: 12,
  },
  fab: {
    backgroundColor: '#007E05',
    width: 56,
    height: 56,
    borderRadius: 28,
    justifyContent: 'center',
    alignItems: 'center',
    elevation: 6,
  },
  fabNearby: { backgroundColor: '#007E05' },
  fabNearbyActive: { backgroundColor: '#FF6B6B' },
  fabBadge: {
    position: 'absolute',
    top: 8,
    right: 8,
    width: 10,
    height: 10,
```

```
    borderRadius: 5,
    backgroundColor: 'white',
  },
});

export default Home;
```

APPENDIX F

(Publishable Paper (IEEE format))

GEORABIES: MOBILE ANDROID-BASED ANIMAL BITE REPORTING AND VACCINATION LOCATOR WITH INTEGRATION OF GIS IN LOS BAÑOS

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***Abstract*— Rabies cases are still one of the public health concerns because of the deadly virus that can infect the human brain through bites from a rabid animal, and the lack of accessible reporting tools that causes underreporting of animal bite cases. It is important to make a decision and take action if a system is in place to monitor animal cases for the community's safety and to raise awareness. The GeoRabies application aims to assist users in reporting animal bite incidents by providing basic information about the incident, such as name, age, date of the incident, type of animal (dog, cat, other), animal category (stray, wild, pet), and locating nearby clinics that are available in the community. It integrates a Geographic Information System (GIS) for mapping and clinic visualization, and enables users to identify available and nearest clinics for rabies vaccination. The researchers used the ISO 25010 software quality model for system evaluation. The app includes animal bite reporting, a mapping module, a clinic locator, an animal bite report generator, and an admin dashboard. Findings revealed that the system is perceived to meet the essential software quality characteristics defined under the ISO 25010 framework to a great extent. High mean scores across most quality attributes indicate that the system provides effective, reliable, and supported rabies prevention efforts.**

Keywords—Geographic Information System (GIS), ISO 25010, Mobile-Based Android Application, Animal Bite Reporting, Vaccination Locator, Incident management

I. INTRODUCTION

Mobile applications are increasingly used in the current digital era to address problems in public safety, healthcare, and education. Many disadvantaged people's inability to access high-quality information limits their opportunities. The lack of easily accessible vaccination programs in rural regions and the delayed reporting of animal bite incidents are two of the main obstacles to rabies control. Many victims are not aware of the correct post-exposure procedures, which frequently leads to avoidable deaths. Global public health is still at risk from rabies, a viral zoonotic illness that causes progressive and deadly inflammation of the brain and spinal cord

An estimated 59,000 people die from rabies each year across more than 150 countries, with 95% of incidents taking place in Asia and Africa. This sure is likely a significant underestimate due to underreporting and imprecise estimates. The impoverished in rural areas bear a disproportionate amount of the disease's burden, with children under the age of 15 accounting for about half of all cases. (World Health Organization - WHO).

The poor in Asia and Africa, the two continents with the highest global rabies burden, most benefit from elimination. The well-known shortage of reliable human and animal surveillance data now hinders the effectiveness of rabies prevention and control campaigns. (Lechenne et al. 2021). Given the high number of rabies-related deaths in Asia compared to other regions, it is determined that this region is the most affected by the virus. As a result, it indicates that there are a lot of rabid animal in Asia, and it is recommended that the WHO, governments, and other organizations carry out their duties and implement certain measures to stop this virus. (Bilal A.)

The purpose of this project is to create an Android mobile application for reporting animal bites and locating rabies vaccinations. Users can find local immunization facilities, report bite events, and access information on rabies treatment and prevention. Using mobile technology, this approach aims to reduce rabies-related deaths by ensuring prompt reporting and access to vaccinations.

The goal of this project is to help people living in communities, especially those in areas with the highest rates of rabies exposure and the fewest healthcare options. Additionally, it

facilitates incident reporting and the finding of neighboring clinics for healthcare practitioners.

Project Objective

The main objective of this research is to develop the GeoRabies mobile application, an Android-based animal bite reporting for animal bite victims and vaccination locator that helps victims navigate to the nearby clinics with the integration of GIS in Los baños. This research specifically aims to:

1. Develop a GIS-integrated mobile application which allows users to locate nearby clinics offering bite-related treatments and access direction.
2. Create a centralized reporting system that enables users to report animal bite incidents.
3. Evaluate the developed system using ISO 25010 and conduct a user acceptance evaluation based on the technology Acceptance Model (TAM)

II. METHODOLOGY

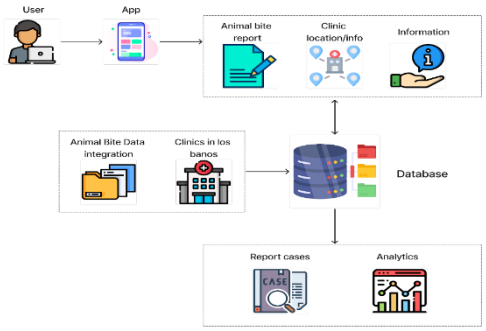


Figure 1. Conceptual Framework

Figure 1 illustrates how users interact with the GeoRabies system to report animal bites and locate vaccination clinics. The system is designed to streamline case reporting, identify high-risk areas, and improve access to vaccination facilities. Users access the GeoRabies mobile application, which provides key functionalities such as Animal Bite Report Submission, Clinic Location Information, and Rabies and Guidelines. Furthermore, the collected data is stored in a centralized database, which integrates additional information, such as animal bite records from the LGU and clinic details (location, name, address, etc.). The system processes the data and provides key outputs, including case reports and analytics for monitoring trends. This allows health authorities to make informed decisions regarding rabies prevention and control efforts. Through this framework, the GeoRabies system enhances public health response and ensures efficient management of animal bite incidents.

Project Design

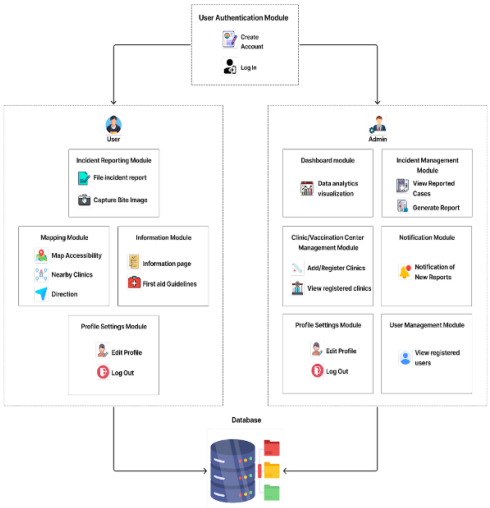


Figure 2. System Architecture

Figure 2 presents the system architecture of the proposed application, which consists of interconnected modules for both users and administrators. The system supports key functions such as user authentication, animal-bite incident reporting, mapping and navigation, information access, clinic and vaccination center management, data analytics, notifications, and user administration. Users must first register and log in through the authentication module to access system features based on their assigned roles. On the user side, the system enables individuals to submit animal-bite incident reports, locate nearby clinics or vaccination centers with navigational guidance, access informative content including first aid guidelines for animal bites, and manage personal profile settings.

On the administrator side, the system provides comprehensive management tools, including a dashboard for monitoring reported cases and trends, incident management for reviewing reports and generating official records, clinic and vaccination center management for maintaining facility data, notifications for newly submitted incidents, and user management for overseeing registered users. Administrators may also update their profile information through the profile settings module. All modules are integrated with a centralized database to ensure efficient data storage and retrieval. This architecture supports seamless system operation, enabling timely incident response, accurate data management, effective coordination with clinics and local health units, and informed decision-making. Overall, the system plays a crucial role in animal-bite case management and supports broader efforts in rabies prevention and control.



Figure 3. Agile Scrum Methodology

Figure 3 illustrates the Agile Scrum Methodology, a dynamic, iterative project management approach that enhances flexibility, teamwork, and ongoing system enhancement. This approach ensures that important functionalities are implemented effectively while maintaining a user-focused design in the creation of GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with GIS Integration in Los Baños. The project uses a sequence of two to four-week sprints to methodically develop and improve the system. Every sprint focuses on delivering the features that enable iterative improvements based on testing and user input. Flexibility, effective work on management, and user needs responsiveness are guaranteed by this methodical approach.

System Flow Chart

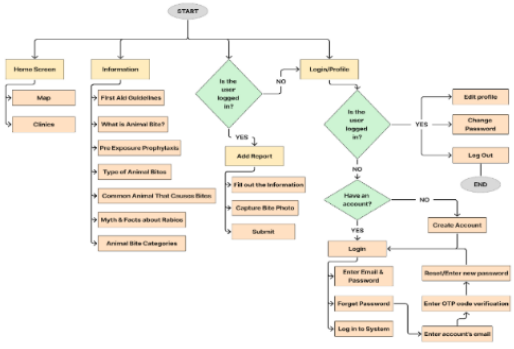


Figure 4. System Flowchart (User)

Figure 4 shows the user-side system flowchart, which defines the system functionalities. First, the user launches the app and navigates to the home screen, where they can use and view the map. Users can see clinics on the map and use the Nearby Clinics Button to display nearby clinics at the current user's location.

Next, in the Information Tab, users can view and read the first-aid guidelines for animal bites. Moreover, learn some facts about animal rabies. What is an animal bite? Pre-Exposure Prophylaxis, Types of animal bites, Common animals that cause bites, Animal Bite Categories, and Myth & Facts about rabies, which can enlighten users who misunderstand rabies. Then, To add a report, The users have to log in to access and fill up the needed information such as Name, Age, Gender, Contact number, Barangay (they live in), Date (of the bite accident), Animal category (stray, pet, wild), Animal type (dog, cat, etc.), Choose bite location to point what barangay area where the user have bitten by an animal.

By entering their registered email address and password users can safely access the system and manage their accounts. After logging in the users can check their account profile, modify personal data, reset their password, and exit the program whenever necessary. Users have complete control over their personal information and account security thanks to the account management tool the users must provide basic details such name, age, gender, phone number, barangay of residence, email address, and password in order to create an account. Accurate identification and efficient communication are made possible by this information the system offers a password recovery tool that asks users to provide their registered email address in the event that they forget their password. After that, users receive an email with a One-Time Password (OTP) that enables them to safely reset their password and get back into their account.

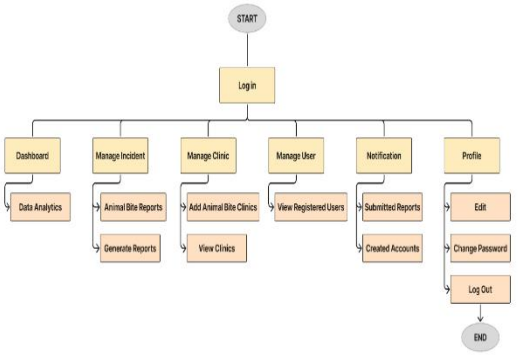


Figure 5. System Flowchart (Admin)

Figure 5 shows the admin-side system flowchart, which defines the system functionalities. First, the admin dashboard serves as the central interface for managing the system. Here are the data analytics for visualization, in which the researchers use different data visualizations, such as bar graphs that represent the total bite report by year and quarter, Age and Gender, and per barangay. The donut graph is also used to determine the animal category (stray, Pet, Wild) and the animal type (dog, cat, other). This

visualizes the risk area for animal bites based on data and user reports. Then, Manage Incidents reports cases, from which the admin can view users' reports and generate reports. After that, the management clinic views registered clinics, including clinic information (Name, Address, Opening Schedule). Also, the Manage user allows the admin to view the user's credentials. The notification enables the admin to be notified about updates, such as new reports and newly created user accounts. Finally, the Admin profile is where the admin can edit their name, email, and password. Once the session is done, the admin can log out, which is also located in their profile.

Project Testing and Evaluation

The researchers adopted the ISO 25010 Software Product Quality Model as the evaluation standard to assess the quality and performance of the developed system, GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with Integration of GIS in Los Baños. This model was selected to ensure that the system meets the desired quality attributes expected from a modern, user-centric mobile application.



Figure 6. Evaluation Procedure

Figure 6 shows the Software Product Quality characteristics from ISO 25010. This model provides a comprehensive framework for evaluating a software product by categorizing quality into several characteristics and sub characteristics such as Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, Flexibility, and Safety.

Population and Data Collection

Table 1 Shows the population of the study consist of residence of Los Baños, Laguna who are people that are usually prone in animal bite incident in their barangays and also IT experts who also included to the study for evaluation and test the system. The researchers used a nonprobability sampling method, a sampling test technique, selecting residents from the surrounding area to participate in the evaluation based on factors or research criteria. The study has a total of 50 respondents and two 3 IT experts who evaluate the system application.

Table 1. Respondents of the Research Project

Category of Respondents	Sample
User	50
IT Experts	3
Total	53

III. RESULTS

System Development and Feature

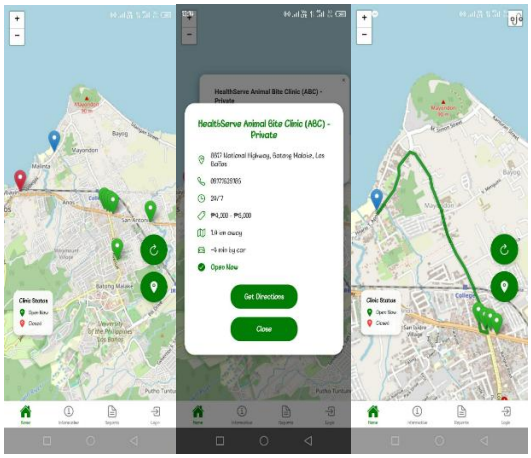


Figure 7. Mapping Module Feature

Figure 7 illustrates the GIS-based clinic mapping module, which allows users to view their current location and locate nearby clinics that provide animal-bite treatment. The module displays essential clinic details such as availability, address, contact information, price range, estimated travel time, and service offerings through interactive map markers. It also includes features for identifying the nearest clinics, providing route directions from the user's location, and indicating clinic availability through a legend. This visual and interactive map interface enhances user experience, particularly during emergencies, by enabling faster access to appropriate medical facilities and supporting timely decision-making following animal-bite incidents.

System Testing

Table 2. Overall TAM Evaluation Results

ITEM	MEAN	SD	INTERPRETATION
A. Perceived Usefulness	3.53	0.02	Strongly Agree
B. Perceived Ease of Use	3.50	0.03	Strongly Agree
C. Personal Innovation	3.37	0.18	Agree
OVERALL WEIGHTED MEAN	**3.47**	**0.09**	**Strongly Agree**

Table 2 indicates that the system achieved an overall weighted mean of 3.47, interpreted as "Strongly Agree," reflecting generally positive user perceptions. High mean scores for Perceived Usefulness (3.53) and Perceived Ease of Use (3.50) suggest that users find the system effective in supporting healthcare information access and animal-bite reporting, as well as easy to learn and use. The results also show favorable user openness to adopting the system, indicating strong usability, acceptance, and readiness for practical implementation.

Table 3. Overall ISO25010 Results

ITEM	MEAN	SD	INTERPRETATION
Functional Suitability	4.00	0.00	Strongly Agree
Performance Efficiency	3.45	0.77	Strongly Agree
Compatibility	3.00	0.67	Agree
Interaction Capability	4.00	0.00	Strongly Agree
Reliability	3.67	0.29	Strongly Agree
Security	3.34	0.73	Strongly Agree
Maintainability	3.73	0.48	Strongly Agree
Flexibility	3.92	0.29	Strongly Agree
Safety	3.74	0.26	Strongly Agree
OVERALL WEIGHTED MEAN	**3.65**	**0.29**	**Strongly Agree**

Table 3 presents the ISO 25010 evaluation results assessed by IT experts, showing an overall weighted mean of 3.65 with a standard deviation of 0.29, interpreted as "Strongly Agree." This indicates that the system largely meets the software quality standards defined in the ISO 25010 framework. High ratings across key quality attributes—including functional suitability, performance efficiency, interaction capability, reliability, security, maintainability, flexibility, and safety—demonstrate the system's strong technical performance. Functional Suitability and Interaction Capability achieved perfect mean scores, reflecting effective task support and intuitive use. Although Compatibility received a slightly lower but still positive rating, the overall findings confirm the system's robustness, usability, and readiness for continued deployment.

IV. DISCUSSIONS

The system demonstrated strong support for user tasks, as reflected by a composite mean of 3.53 in the Perceived Usefulness category, interpreted as "Strongly Agree." High ratings indicate that the system provides reliable and relevant information that enhances decision-making accuracy, resource efficiency, and overall productivity. The positive evaluation of the machine learning integration further confirms user satisfaction with the system's advanced capabilities, validating that the system effectively meets the functional needs of its intended users.

The system achieved a composite mean of 3.50, interpreted as "Strongly Agree," indicating that users found the application intuitive and easy to use. High ratings for interface design and clarity of system functions show minimal difficulty in navigation and interaction. Additionally, the ability to complete tasks efficiently and recall system functions over time demonstrates that the interface supports both regular and occasional users, contributing to improved accessibility and user acceptance.

The results yielded a composite mean of 3.37, interpreted as "Agree," indicating that users generally demonstrate a positive attitude toward adopting new technologies. High ratings for willingness to learn new systems and openness to innovative platforms reflect strong user receptiveness. Although experimentation with new technologies received a slightly lower rating, the overall findings suggest adequate technological readiness, supporting the system's potential for wider adoption.

This project focused on developing an Android-based mobile incident reporting system

for animal bites that integrates GIS, specifically in the Los Baños area. It aims to provide users who have been bitten by an animal with directions to seek immediate treatment and first aid. The researchers also developed an informative interface on rabies, including proper steps to take in case of exposure, to raise awareness and improve accessibility for potential users. The geographic area of the system application is restricted to public clinics, hospital clinics, private clinics, and any other establishments that provide care for animal bites in Los Baños, Laguna. On the contrary, there may be situations in which private clinics are unwilling to provide the researchers with their data or information for the project. In addition, some system functionalities may impose restrictions due to the requirement to use a billing account.

V. CONCLUSION

The mobile application integrates GIS using OpenStreetMap to help users locate nearby clinics that provide animal-bite treatment and access navigation directions. A centralized reporting system was developed to allow users to submit animal-bite reports and identify nearby clinics for immediate medical assistance, supporting both user needs and administrative incident management.

The system was evaluated using the ISO/IEC 25010 software quality model, covering key attributes such as functional suitability, performance efficiency, compatibility, reliability, security, maintainability, adaptability, and safety. In addition, the Technology Acceptance Model (TAM) was applied to assess user acceptance in terms of perceived usefulness, ease of use, and personal innovation. Results from both evaluations indicate that the GeoRabies Application System is reliable, user-friendly, and compliant with international quality standards, confirming its readiness for real-world deployment.

Recommendation

Based on the results and findings of the study, the researchers propose the following recommendations for future researchers who wish to enhance, expand, or further develop the GeoRabies: Mobile Android-Based Animal Bite Reporting and Vaccination Locator with Integration of GIS in Los Baños:

Future enhancements to the GeoRabies system include integrating Google account sign-up and login to simplify user authentication and improve accessibility. Expanding the application's compatibility beyond Android—such as developing versions for iOS and other platforms—would allow broader user adoption. Additionally, enabling full offline navigation and implementing real-time location tracking would improve route guidance, particularly in low-connectivity areas and emergency situations.

It is also recommended to develop a web-based administrative dashboard to enhance efficiency in system management and data handling. Finally, establishing a regular system maintenance framework, including updates, performance optimization, bug fixes, and security improvements, is advised to ensure long-term reliability and continued effectiveness of the application.

REFERENCES

[1] Asif Bilal, (2017) Rabies is a Zoonotic Disease: A Literature Review https://www.researchgate.net/publication/358802264_Rabies_is_a_Zoonotic_Disease_A_Literature_Review

[2] Ahmad N, Nawi AM, Jamhari MN, Nurumal SR, Mansor J, Zamzuri M'IA, Yin TL, Hassan MR. (2022) Post-Exposure Prophylactic Vaccination against Rabies: A Systematic Review. Iran J Public Health. 967-977. doi: 10.18502/ijph.v51i5.9412. PMID: 36407746; PMCID: PMC9643235. https://pmc.ncbi.nlm.nih.gov/articles/PMC9643235/

[3] B. B. Mennecke, M. D. Crossland, (1996) Geographic Information Systems: Applications and Research Opportunities for Information Systems Researchers. https://www.researchgate.net/publication/3624069_Geographic_information_systems_applications_and_research_opportunities_for_information_systems_researchers

[4] Chandi, L., Silva, C., Martínez, D., & Gualotuña, T. (2017). Mobile application development process: A practical experience. Paper presented at the 12th Iberian Conference on Information Systems and Technologies (CISTI), Lisbon, Portugal. https://doi.org/10.23919/CISTI.2017.7975825

[5] Deray, R., Gripon, S., Ulanday, C., Roces, M. C., Amparo, A. C., Attlan, M., Demont, C., Kieffer, A., Miranda, M. E. G., Rivera, C. (2018). Protecting children from rabies with education and pre-exposure prophylaxis: A school-based campaign in El Nido, Palawan, Philippines https://www.researchgate.net/publication/322219384_Protecting_children_from_rabies_with_education_and_pre-exposure_prophylaxis_A_school-based_campaign_in_El_Nido_Palawan_Philippines

[6] Ershad Ali, (2020) Geographic Information System (GIS): Definition, Development, Applications & Components https://www.researchgate.net/publication/340182760_Geographic_Information_System_GIS_Definition_Development_Applications_Components

[7] Estima NM, Wada MY, Rocha SM, Moraes DS, Ohara PM, Vargas A, Assis DM. (2022) Description of human anti-rabies post-exposure prophylaxis care notifications

in Brazil, 2014-2019. Epidemiol Serv Saude. e2021627. doi: 10.1590/S2237-96222022000200002. PMID: 35730814; PMCID: PMC9887991. https://pmc.ncbi.nlm.nih.gov/articles/PMC9887991/#sec9

[8] G.R. Cabrera, M.C. S. Guillermo, S. Evangelio , E.O. Diamante, (2024) A Centralized Mobile App for Dog Rabies Vaccination Reporting by Private and City Veterinarians in Davao City: Issue Joint Conference & Symposium on Health

[9] Guiquing, F. L. (2021). Animal bite mapping (ABMAP) using mobile apps. Psychology and Education Journal, 58(2), 3398–3407. https://psychologyandeducation.net/pae/index.php/pae/article/view/3398

[10] Hanna, K. T., & Wigmore, I. (n.d.). Mobile app. TechTarget. Retrieved from https://www.techtarget.com/whatis/definition/mobile-app

[11] I. Amaral, (2014) In book: Encyclopedia of Social Media and Politics (pp.574-575) Chapter: Geotagging Publisher: Sage Publications Editors: Kerric Harvey https://www.researchgate.net/publication/270887898_Geotagging

[12] Kessels JA, Recuenco S, Navarro-Vela AM, Deray R, Vigilato M, Ertl H, Durrheim D, Rees H, Nel LH, Abela-Ridder B, Briggs D. (2016) Pre-exposure rabies prophylaxis: a systematic review. Bull World Health Organ. 2017 Mar 1;95(3):210-219C. doi: 10.2471/BLT.16.173039. PMID: 28250534; PMCID: PMC5328107. https://pmc.ncbi.nlm.nih.gov/articles/PMC5328107/

[13] Koury R, Warrington SJ. (2022) Rabies In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK448076/

[14] L. Leonardo, BA. Crisostomo, JA. Solon, P. T. Rivera, A. Marcelo, J. Villasper, (2007) Geographical information systems in health research and services delivery in the Philippines Geospatial Health 1(2):147-55 DOI:10.4081/gh.2007.263 SourcePubMed LicenseCC BY-NC 4.0 https://www.researchgate.net/publication/23155117_Geographical_information_systems_in_health_research_and_services_delivery_in_the_Philippines

[15] M.M. Nowak, K. Słupecka, B. Jackowiak, (2021) Geotagging of natural history collections for reuse in environmental research, Ecological Indicators, Volume 131,108131,ISSN 1470-160X, https://doi.org/10.1016/j.ecolind.2021.108131.

[16] Maniscalco K, Edens MA. (2022) Animal Bites. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK430852/

[17] Nadal, D., Bote, K., Masthi, R., Narayana, A., Ross, Y., Wallace, R., & Abela, B. (2023). Rabies post-exposure prophylaxis delivery to ensure treatment efficacy and increase compliance. IJID one health, 1, 100006. https://doi.org/10.1016/j.ijidoh.2023.100006

[18] Preiss, S., Chanthavanich, P., Chen, L. H., Marano, C., Buchy, P., van Hoorn, R., … Mukherjee, P. (2018). Post-exposure prophylaxis (PEP) for rabies with purified chick embryo cell vaccine: a systematic literature review and meta-analysis. Expert Review of Vaccines, 17(6), 525–545. https://doi.org/10.1080/14760584.2018.1473765

[19] Pick, J. (2004). Geographic information systems: A tutorial and introduction. Redlands. Retrieved from https://www.academia.edu/69074884/Geographic_information_systems_A_tutorial_and_introduction

[20] Patrick Soentjens, Mieke Croughs, (2021) Simplified rabies pre-exposure prophylaxis in last-minute travellers, Journal of Travel Medicine, Volume 28, Issue 1, taaa185, https://doi.org/10.1093/jtm/taaa185

[21] Radicelli-García, C., Floril, M., Cevallos, P., & Bodero Poveda, E. (2022). Geographic information systems and their application in the social sciences: A literature review. Journal of Geography and Cartography, 5(1). https://doi.org/10.24294/jgc.v5i1.1679

[22] Rupprecht CE. (1996) Rhabdoviruses: Rabies Virus. In: Baron S, editor. Medical Microbiology. 4th edition. Galveston (TX): University of Texas Medical Branch at Galveston; Chapter 61. Available from: https://www.ncbi.nlm.nih.gov/books/NBK8618/

[23] Salm, M. (2025). Geo-tagging. Better Evaluation. Retrieved from https://www.betterevaluation.org/methods-approaches/methods/geo-tagging

[24] Singh, R., Singh, K. P., Cherian, S., Saminathan, M., Kapoor, S., Manjunatha Reddy, G. B., … Dhama, K. (2017). Rabies – epidemiology, pathogenesis, public health concerns and advances in diagnosis and control: a comprehensive review. Veterinary Quarterly, 37(1), 212–251. https://doi.org/10.1080/01652176.2017.1343516

[25] Smith, C., Ortal, A., Girasol, M., Suzuki, S., & Coughlan, C. (2024). Rabies in the Philippines: A call to action. The Lancet Regional Health – Western Pacific, 49, 101156. https://www.thelancet.com/journals/lanwpc/article/PIIS2666-6065(24)00150-0/fulltext

[26] Shadani S, Ghasemi F, Kamali M, Navaeifar MR, Alikhani A, Alian S (2022) Rabies Virus Infection: A Case Series and Literature Review. Journal of Pediatrics Review 10(4):315-320. http://dx.doi.org/10.32598/jpr.10.4.369.5

[27] Swathi, M., Sabeena, S., Abdulmajeed, J., Pattanaik, S., Dsa, O., Shubha, H. S., . . . Arunkumar, G. (2020). Development and evaluation of a mobile app for guiding rabies prophylaxis among health-care professionals in India. Indian Journal of Community Medicine, 45(4), 473. https://doi.org/10.4103/ijcm.ijcm_510_19

[28] Schrodt, C. A., Dilius, P., Gibson, A. D., Crowdis, K., Fénelon, N., Ross, Y., Bonaparte, S., Joseph, H. C., & Wallace, R. M. (2023). Electronic application for rabies management improves surveillance, data quality, and investigator experience in Haiti. Frontiers in veterinary science, 10, 1052349. https://doi.org/10.3389/fvets.2023.1052349

[29] T. Dutta, (2014) Rabies: An overview International Journal of Advanced Medical and Health Research 1(2):39 DOI:10.4103/2349-4220.147998 LicenseCC BY-NC-SA 4.0 https://www.researchgate.net/publication/293801831_Rabies_An_overview

[30] Volle, A. (2025, January 23). app. Encyclopedia Britannica. Retrieved from https://www.britannica.com/technology/mobile-app

[31] Wieczorek WF, Delmerico AM. (2009) Geographic Information Systems. Comput Stat. 1(2):167-186. doi: 10.1002/wics.21. PMID: 20717487; PMCID: PMC2921721. https://pmc.ncbi.nlm.nih.gov/articles/PMC2921721/#abstract1

[32] World Health Organization, Rabies (2024) https://www.who.int/news-room/factsheets/detail/rabies

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APPENDIX H

(Grammarly Writing Assistant Report)

CHAPTER 1 - 3

by Ronah joy Ramirez

General metrics

69,670	10,143	767	40 min 34 sec	1 hr 18 min
characters	words	sentences	reading time	speaking time

Score

99

This text scores better than 99% of all texts checked by Grammarly

Writing Issues

1	✓	1
Issues left	Critical	Advanced

Plagiarism

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Chapter 4 - 5

by Ronah joy Ramirez

General metrics

36,523	5,391	363	21 min 33 sec	41 min 28 sec
characters	words	sentences	reading time	speaking time

Score

99

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Writing Issues

1	1	✓
Issues left	Critical	Advanced

Plagiarism

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APPENDIX I

(Curriculum Vitae)



RAGIMAR TOLO

+63960-512-1613

ragimartolo@gmail.com

m.me/ragimar.tolo.12

Camp Gen. Macario Sakay, Lalakay, Los Baños, Laguna

Education

Bachelor of Science in Information Technology
Laguna State Polytechnic Univ.
2022 - Present

Information & Communication Technology
Los Baños Bambang Senior High
2020 - 2022

Skills

Web Design

Basic Computer Trouble Shoot

Basic Programming

Knowledge of IT Related

Language

English

Filipino

Profile

Reliable and hardworking with a sense of genuine commitment to doing the job well. Eager to learn, take initiative, and contribute positively. Known for a strong work ethic, attention to detail, and a professional attitude in all tasks.

Personal Details

Age: 22
Date of Birth: Feb 21 2003
Citizenship: Filipino
Civil Status: Single
Height: 5'1
Weight: 47Kl

Work Experience

2021 – 2022

Canon Business Machine Philippines Inc.
Production Operator

- Working well with the corresponding team.
- Manage to Attain Certificate of Emploment



Abas, Chester Jay

Objective

A dedicated and hardworking individual seeking a role where I can utilize my skills in teamwork, adaptability, and IT-related knowledge to support organizational goals and develop professionally.

Contact

0910-240-4223

chesterjayabas@gmail.com

Pansol, Calamba City
Laguna

Skills

- Web Design
- Basic Programming
- Computer Knowledge (IT Related)
- Flexibility
- Hardworking
- Teamwork

Personal Background

- **Date of Birth:** May 14, 2002
- **Status:** Single
- **Age:** 23
- **Sex:** Male
- **Citizenship:** Filipino
- **Weight:** 50 kg
- **Height:** 5'1

Education

(2022 - Present)

LAGUNA STATE POLYTECHNIC UNIVERSITY

Bachelor of Science in
Information Technology

(2018 - 2020)

SAINTS JOHN AND PAUL EDUCATIONAL FOUNDATION INC.

Academic Strand: ICT

Experience

(2021 -2022)

PRODUCTION OPERATOR

Cam Connectivity Inc.



VINCE LEO POBLETE

vincepoblete0322@gmail.com | 09608540872 | 184 centro lalakay los baños laguna

PROFILE INFO

A motivated individual with a strong willingness to learn and grow in a professional environment. Able to adapt quickly to new situations and handle responsibilities with a positive attitude.

SKILLS

- Problem Solving
- Teamwork
- Time Management
- Leadership
- Effective Communication
- Critical Thinking
- Knowledge of IT Related

LANGUAGES

- English (Basics)
- Filipino (Fluent)

EDUCATION

ELEMENTARY
Lalakay elementary school

HIGH SCHOOL
Pulo national high school

COLLEGE
Laguna State Polytechnic University

EXPERIENCE

2016 – 2019

Production / Spray Painter
Suzuki Philippines

- Applied paint and finishes to motorcycle/vehicle parts following quality standards
- Ensured proper color matching and smooth, even coatings
- Maintained a clean and safe work area while following company procedures